Do ESG Investors Care About Carbon Emissions? Evidence From Securitized Auto Loans*

Christian Kontz

Stanford GSB

April 14, 2026

Abstract

This paper documents that ESG preferences in fixed income markets affect real outcomes through securitization. I find that higher ESG scores lower funding costs for auto loan securitizations and that captive lenders pass through these savings to consumers: a 9 basis points ESG funding advantage translates into a 21 to 33 basis points lower consumer interest rate. The high pass-through is driven by an increased probability of receiving a subsidized interest rate (e.g., 0% financing). Lower consumer borrowing costs increase loan demand by up to 4.9%. Yet because issuer ESG scores are weak proxies for financed CO_2 emissions, the same mechanism reduces the cost of capital for high-emission vehicles. A subjective-beliefs model can explain how investors who intend to price emissions but implement this via ESG scores lower the cost of capital of high-emission vehicles.

JEL Classification: G12, G18, G20, G41, Q56

Keywords: Auto Loan Securitization, Captive Lending, Carbon Emissions, ESG Investing, Greenium, Sustainable Investing

*I thank Joe Aldy (discussant), Hunt Allcott, Jan Bena (discussant), Jonathan Berk, Sebastian Bauer, Elaine Buckberg, Darrell Duffie, Alex Edmans, Tim Eisert (discussant), Lawrence Goulder, Lars Hansen, Sebastian Hanson, Ben Hébert, Mete Kılıç (discussant), Hanno Lustig, Adrien Matray, Marcus Opp, Amit Seru, Lior Shabtai (discussant), Ľuboš Pástor, Monika Piazzesi, Stefan Reichelstein, Edward Watts, Luigi Zingales, as well as seminar and conference participants at the AFA Annual Meeting 2025, FIRS Conference 2024, SoFiE Conference 2024, OU-RFS Climate and Energy Finance Conference 2024, CEPR-ESSEC-Luxembourg Conference on Sustainable Financial Intermediation, Harvard Climate Economics Workshop, UChicago, UC Santa Cruz, USC Marshall PhD Conference 2023, and GEA Winter Meeting 2022 for helpful comments. All errors remain my own. First version: September 27, 2022.

1 Introduction

A central goal of environmental, social, and governance (ESG) investing is to raise the cost of emitting carbon dioxide (CO_2) by redirecting capital toward "green" assets and away from "brown" assets.[1] The hope is that a higher cost of capital for high-emission assets, in turn, reduces demand for those activities and mitigates climate change. However, estimating whether ESG investing successfully raises the cost of emitting CO_2 is challenging: a clean measurement must hold an asset's risk exposure constant while varying its greenness and must quantify the actual environmental impact of the investment.

This paper sheds light on whether ESG investing increases the cost of emitting CO_2 by addressing these challenges. I study the effect of ESG investing on the cost of capital of automobile asset-backed securities (ABS) and its pass-through to consumer interest rates. Auto loan securitizations finance over 20% of vehicle sales and serve as a critical link between consumers and financial markets. By changing the cost of capital for high-emission vehicles, ESG investing could shift consumer demand away from high-emission vehicles toward greener alternatives.[2] The auto ABS market is a natural laboratory for this question: over the past 10 years, issuers, investors, rating agencies, and industry associations have all devoted explicit attention to ESG considerations in this market (see Appendix Figure A1).

Auto loan securitizations provide an ideal setting to study the impact of ESG investing on equilibrium asset prices and evaluate its real effects. The pooling and tranching of auto loans create highly liquid securities with specified risk exposures (DeMarzo, 2005). I exploit the safe asset nature of senior tranches and use variables derived from loan-level data to hold risk exposure constant across securities. I then test whether the greenness of a securitization influences its cost of capital. I contrast two measures of greenness: (i) ESG scores, commonly used by the asset management industry, and (ii) the collateral pool's CO_2 emissions. By comparing the influence of ESG scores and CO_2 emissions, I test the often implicit assumption that a "green" premium is associated with a higher cost of emitting CO_2. Finally, I use granular loan-level data to estimate the pass-through of ESG investing to consumer interest rates and calculate the corresponding shift in consumer auto loan demand. This allows me to evaluate ESG investing's potential to shift consumer demand away from high-emission vehicles and toward greener alternatives.

[1] I use "ESG investing" as a shorthand for all investment strategies aimed at addressing environmental externalities. Asset managers rank climate change, CO_2 emissions, and fossil fuel divestment as the top ESG criteria (USSIF, 2022).

[2] Reducing vehicle emissions is a key policy goal to mitigate climate change. Transportation accounts for more than 35% of energy-related CO_2 emissions in the U.S. and became its leading source in 2017 (CBO, 2022).

I collect data on 17.8 million vehicle loans that serve as collateral for all auto ABS issued from 2017 to 2022 by captive lenders of manufacturers, banks, non-bank finance companies, and retailers. I estimate the lifetime CO_2 emissions of each vehicle loan by merging collateral data at the make, model, and year level with CO_2 emissions data from the Environmental Protection Agency (EPA). This enables me to calculate the financed CO_2 emissions of each securitization and to quantify its environmental impact. Additionally, I collect issuer-level ESG scores from MSCI, Sustainalytics, Refinitiv, and S&P to analyze how ESG scores affect the cost of capital of auto ABS.

I start by documenting large cross-sectional differences in CO_2 emissions across auto ABS. For instance, auto ABS issued by Ally Bank finance an average of $55tCO_2$ per vehicle, while those issued by Ford Credit finance an average of $95tCO_2$ per vehicle. The large cross-sectional differences in CO_2 emissions imply that, much like motorists choose between high- and low-emission vehicles, investors have the choice between securities that finance high- and low-emission vehicles. I exploit the fact that auto ABS have large environmental differences but low and similar levels of risk exposure to test whether CO_2 emissions or ESG scores influence their cost of capital.

The ESG scores and environmental pillar scores of auto ABS issuers do not capture the large differences in CO_2 emissions. Both scores correlate positively with security-level CO_2 emissions. Decomposing the variance of ESG scores and CO_2 emissions shows that ESG and environmental scores vary significantly less across issuers than the collateral pool's emissions. The smaller variance and positive correlation with CO_2 emissions make ESG and environmental scores poor proxies for the actual environmental impact of auto ABS. The reasons for this discrepancy are that (a) environmental scores of banks and non-bank finance companies do not reflect the CO_2 emissions content of the loans they securitize, and (b) environmental scores among vehicle manufacturers often do not capture CO_2 emissions from the usage of sold products (Scope 3 Category 11). However, emissions from the usage of sold vehicles are up to 1,000% larger than production emissions (Scope 1, 2, and 3 upstream).[3] Consequently, investors who rely on ESG scores to allocate capital may inadvertently subsidize CO_2 emissions. I use this fact to test whether a green premium based on ESG scores actually increases the cost of emitting CO_2.[4]

I next test whether ESG investing impacts equilibrium prices and quantities in the auto ABS market. I motivate my identification strategy with a stylized asset pricing model that features a green

[3] For example, the utilization emissions of a Volkswagen Passat are 527% larger at $47tCO_2$ than its production emissions of $7.5tCO_2$ (Buberger, Kersten, Kuder, Eckerle, Weyh, and Thiringer, 2022). Nissan (2024, p. 24) reports that its Scope 3 Category 11 emissions are 747% larger than its combined Scope 1, 2, and 3 upstream emissions.

[4] Amel-Zadeh and Serafeim (2018) report that 82% of surveyed investment firms, accounting for 43% of global institutional assets under management (AUM), consider ESG information when making investment decisions. More than 66% of their survey respondents use ESG information to screen investment opportunities or tilt their portfolios.

convenience yield in the spirit of Krishnamurthy and Vissing-Jorgensen (2012). The identification strategy allows me to answer the following questions: once an investor is certain that the loan will be repaid as agreed, does the loan's greenness influence the investor's decision, and is this reflected in equilibrium asset prices?

Two features of the securitization market allow me to rule out confounders in my effort to identify the causal effect of ESG investing. First, auto ABS are highly standardized debt instruments. Their senior tranches are considered safe assets similar to U.S. Treasurys (Gorton, 2017). Only a few parameters distinguish auto ABS beyond their collateral pools. I exploit the safe asset nature of senior tranches and use variables derived from loan-level data to control for any remaining differences across securities. Second, the security design of auto ABS reduces the number of risk factors. The main risk factor for AAA-rated senior tranches of auto ABS is prepayment. Consumer and loan characteristics determine prepayment risk rather than the issuer identity or the collateral. Borrowers with high interest rate loans prepay when interest rates fall, regardless of the ESG score of the issuer or the CO_2 emissions of the collateral they finance.[5] Moreover, the granularity of the loan-level data allows me to include variables that capture ex-post prepayments as placebo controls. This alleviates concerns that investors use greenness to infer risky payoffs, as in Pedersen, Fitzgibbons, and Pomorski (2021), rather than to express their non-pecuniary preferences. Controlling for both predictors and ex-post performance together with fixed effects that capture security design removes as much unobserved heterogeneity as possible.

I find that issuers with high ESG or environmental pillar scores have significantly lower issuance spreads. Moving from the 20th to the 80th percentile of ESG scores reduces spreads by 9 bps (0.3 standard deviations), holding security design and risk exposure constant. However, I also find that *high-emission* collateral pools have a *lower* cost of capital. Moving from the 20th to the 80th percentile of tCO_2 per USD (e.g., moving from Toyota to Ford) reduces issuance spreads by 3 bps, a 7% decrease relative to the mean issuance spread.

ESG scores, rather than CO_2 emissions, are associated with investment decisions and correlate with the cost of capital of auto ABS. A horse race between CO_2 emissions and ESG scores shows that ESG scores dominate emissions in explaining the cost of capital. The CO_2 coefficients shrink toward zero and lose statistical significance when ESG scores are included in the model. Coefficients on ESG scores, however, remain stable and statistically significant. This suggests that investors rely on ESG scores rather than actual CO_2 emissions to evaluate the environmental impact of auto ABS. However, the reliance on ESG scores leads to a subsidy for high-emission auto

[5] I directly test this in Table 4 and find no evidence that ESG scores or the collateral's emissions affect prepayment or default rates.

ABS since ESG scores and CO_2 emissions correlate positively.

I translate the observed differences in issuance spreads into an ESG convenience yield, motivated by my ESG asset pricing model. This ESG convenience yield provides seigniorage to issuers of ESG assets and lowers their borrowing cost. The ESG convenience yield rose from 0.02% in 2018 to 0.50% in 2022. The average ESG convenience yield over the sample period is 0.39% p.a., comparable to estimates documented in equity markets. For instance, Avramov, Lioui, Liu, and Tarelli (2024) estimate an ESG convenience yield for stocks ranging from 0.37% to 0.66%, while Eskildsen, Ibert, Jensen, and Pedersen (2024) report a convenience yield of 0.5% per year.

Concerns about climate change correlate with the pricing of the ESG convenience yield into issuance spreads. I test whether concerns about climate change are associated with the pricing of ESG scores by interacting ESG scores with the monthly Media Climate Change Concerns Indices of Ardia, Bluteau, Boudt, and Inghelbrecht (2023). I find evidence that media attention about the environmental impact of climate change and climate change in general significantly influences ESG score pricing. In contrast, using both the car-industry transition risk index and the general transition risk index, I find no evidence that concerns about transition risks affect ESG score pricing. These results lend support to the hypothesis that non-pecuniary preferences for ESG investing are shaped by environmental concerns about climate change.

Banks that have publicly committed to align their portfolios with the Paris Agreement intermediate the pricing of ESG preferences. Auto ABS reach investors through a small set of underwriting banks, some of which have made commitments through the Science Based Targets initiative (SBTi) to decarbonize. The identity of the lead underwriters varies across deals issued by the same issuer within the same year, which allows me to isolate the effect of underwriter climate commitments from issuer and collateral characteristics. I find that deals placed by at least one SBTi-committed underwriter exhibit larger ESG premium. SBTi-committed underwriters capture part of this premium through higher underwriting discounts charged to issuers. Both the pricing of ESG scores and underwriting discounts are larger in deals placed by SBTi-committed underwriters when media attention to climate change is high. Relative to Kacperczyk and Peydró (2021) and Sastry, Verner, and Marques-Ibanez (2024), who study the effect of banks' climate commitments on balance-sheet lending, I document that banks transmit ESG preferences through the securitization channel in which they act as intermediaries rather than balance-sheet lenders.

The more than $1.1 trillion in net flows into ESG funds over the past decade (van der Beck, 2026) similarly correlate with the pricing of ESG scores into issuance spreads.[6] By interacting ESG

[6] Pastor, Stambaugh, and Taylor (2025) estimate that institutional ESG tilts grew from 17% to 27% over that time.

scores with flows into ESG funds, I find that a $100bn increase in net flows to ESG funds is associated with a 2 bps reduction in issuance spreads. I directly examine the portfolios of ESG mutual funds to test whether CO_2 emissions or ESG scores influence their portfolio choice. ESG funds invest more in auto ABS from issuers with high ESG scores than non-ESG funds. However, the portfolio analysis also shows that ESG funds hold higher portfolio shares in high-emission auto ABS than non-ESG funds. ESG funds allocate approximately 20.2% less capital to auto ABS with emissions below the median than non-ESG funds do. These findings are difficult to reconcile with common ESG strategies that prescribe outright exclusion or best-in-class investment—at least under standard assumptions about investor beliefs.[7]

The results above suggest that ESG investors use ESG scores to express their non-pecuniary preferences over greenness but inadvertently subsidize CO_2 emissions due to a positive correlation between ESG scores and CO_2 emissions. This need not mean subsidizing emissions is the intended effect: investors may instead intend to raise the cost of emitting CO_2 but hold imperfect information about the emissions of the collateral they finance. To separate intent from outcome, I develop a stylized model of ESG investors' subjective—and potentially biased—beliefs about CO_2 emissions in auto ABS. Emissions enter as latent variables, and investors form beliefs about them on the basis of ESG scores.[8] Combining the model with survey evidence from Haber, Kepler, Larcker, Seru, and Tayan (2022) allows me to recover an estimate of the implied subjective beliefs. Alternatively, assuming investors apply a representativeness heuristic (Gennaioli and Shleifer, 2010) to infer the score–emissions relationship recovers the intended price of CO_2 emissions in auto ABS. Both approaches yield similar results. A simple calibration of model implies that ESG investors intended to demand 0.2% higher issuance spreads for a 1% higher CO_2 emissions intensity. Their subjective beliefs about the relationship between ESG score and CO_2 emissions, however, diverge sharply from reality: they imply that a 1% higher ESG score corresponds to a 0.2% decrease in emissions intensity, while the true relationship is a 0.1% *increase*.

Lastly, I test whether the ESG convenience yield affects consumer auto loan demand. The integration of the consumer loan market and financial markets via securitization provides an opportunity to study whether issuers who benefit from the ESG convenience yield pass on lower borrowing costs to consumers. The impact of ESG investing on consumer loan demand depends on the pass-

[7] There is a long-running debate about which strategy ESG investors should follow: exit (exclusion) or voice (activism) (Hirschman, 1970). Broccardo, Hart, and Zingales (2022) analyze the relative effectiveness of these strategies, while Edmans, Levit, and Schneemeier (2022) examine whether exclusion or best-in-class investment is more effective.

[8] Informal conversations with market participants confirm that investors did not have real-time access to collateral-pool CO_2 emissions data until the end of my sample period. Appendix A.4 shows that even dedicated ESG mutual funds relied on ESG scores to assess the greenness of auto ABS.

through of issuance spreads in the ABS market to consumer rates and on the price elasticity of consumer loan demand with respect to these rates. I estimate the endogenous pass-through elasticity using instrumental variables that isolate exogenous variation in common funding costs of auto loan lenders. I use price elasticity estimates from the literature (e.g., Argyle, Nadauld, and Palmer, 2020). With estimates of pass-through and price elasticity in hand, I translate the 9 bps difference in issuance spreads into a percentage change in consumer loan demand.

Consumers financing with captive lenders benefit from the ESG convenience yield through lower borrowing costs. Estimates of the pass-through elasticity imply that a 9 bps decrease in auto ABS spreads translates into a 21 bps to 33 bps lower consumer rate due to the non-linear effect of loan subsidies. Manufacturers often use the vertical integration of credit provision to increase their sales by offering subsidized interest rates to consumers (Benetton, Mayordomo, and Paravisini, 2021).[9] The resulting changes in individual consumer loan demand range from 1.1% to 4.9%. This translates into an increase in demand of $374 to $1,634 for the average loan in my sample.

Buyers of high-emission vehicles using captive lender financing show the strongest association between ABS funding conditions and borrowing terms. Lower ABS issuance spreads are systematically related to both lower consumer interest rates and a higher probability of receiving subsidized financing, concentrated almost entirely among the pickup truck segment of captive lenders with high ESG scores. Relative to passenger car buyers, the 9 bps ESG basis spread corresponds to an estimated 59 bps to 65 bps reduction in interest rates and an 20 p.p. to 23 p.p. increase in subsidy probability for truck buyers who finance their vehicle through a captive lender with a one standard deviation higher ESG score. These findings indicate that buyers of high-emission vehicles benefit the most from the ESG convenience yield through lower consumer interest rates and an increased likelihood of receiving subsidized financing when borrowing from captive lenders.

In summary, auto ABS issuers with high ESG scores who securitize high-emission vehicles enjoy a lower cost of capital. The lower cost of capital for high-emission ABS is unrelated to risk. Rather, heightened concerns about climate change and large capital flows into ESG funds over the past decade seem to drive this lower cost of capital. ESG mutual funds allocate more capital to auto ABS of high-ESG issuers even when those issuers finance high-emission vehicles. A model of subjective beliefs in which investors heuristically infer CO_2 emissions from ESG scores can explain these findings. Consumers financing high-emission vehicles through captive lenders benefit from the ESG convenience yield through lower borrowing costs. The results are robust across various measures of greenness, samples, specifications, and estimators.

[9] An 9 bps lower issuance spread increases the probability of receiving a subsidized rate (e.g., 0% APR) from a captive lender by 7 p.p. The expected decrease in the consumer interest rate conditional on subsidization is 28 bps.

The paper is organized as follows. The remainder of the introduction discusses the related literature and contribution. Section 2 describes the data. Section 3 provides an overview of the auto ABS market. Section 4 develops a stylized green asset pricing model, discusses identification, and estimates the effect of ESG investors on the cost of capital. Section 5 estimates the pass-through of the ESG convenience yield to consumer interest rates and the implied effect on loan demand. Section 6 provides a discussion of the findings. Section 7 concludes.

Related Literature The rise of ESG investing spurred extensive research.[10] Theoretical studies show that if ESG investors comprise a significant share of the market, green assets will have a lower cost of capital. Heinkel, Kraus, and Zechner (2001) model an equilibrium in which ESG investors increase the cost of capital for polluting firms. Oehmke and Opp (2024) outline conditions under which ESG investors affect firm behavior, considering social costs and financing constraints. Pástor, Stambaugh, and Taylor (2021) examine how changes in ESG preferences impact asset prices. Berk and van Binsbergen (2025) study equity divestment in a single-period mean-variance model. I contribute by proposing a stylized asset pricing model featuring a green convenience yield in the spirit of Krishnamurthy and Vissing-Jorgensen (2012). I further highlight how incorrect subjective beliefs about the efficacy of ESG scores can lead to the unintended consequence that brown securities have a lower cost of capital.

This paper introduces several innovations to the empirical literature and is the first to study the effects of environmental externalities, ESG scores, and ESG investing on the pricing and holdings of asset-backed securities. My findings relate to other studies on the green premium in debt markets, such as Pástor, Stambaugh, and Taylor (2022), who report a 5 bps lower yield for Germany's green Bunds, and Baker, Bergstresser, Serafeim, and Wurgler (2022), who estimate a 6 bps green premium in U.S. municipal and corporate bonds.[11] However, my results reveal a tension between ESG investors' goals and the use of misleading issuer-level ESG scores. I find that ESG investing can have a meaningful impact, but it does not necessarily increase the cost of emitting CO_2.[12]

This paper also relates to a growing literature on the measurement and economic consequences of ESG ratings and climate risk. Berg, Koelbel, and Rigobon (2022b) document substantial divergence across ESG rating providers, driven primarily by differences in scope and measurement, while Berg, Fabisik, and Sautner (2020) show that providers retroactively revise historical scores.

[10] See Gillan, Koch, and Starks (2021) and Hong and Shore (2023) for excellent reviews.

[11] See also Goss and Roberts (2011), Chava (2014), Zerbib (2019), Flammer (2021), Seltzer, Starks, and Zhu (2026), Aswani and Rajgopal (2025).

[12] Relatedly, Hartzmark and Shue (2026) argue that redirecting capital from brown to green companies may backfire due to limited improvement potential in green firms and deterioration in brown firms. In contrast, I focus on vehicle loans for which adjusting the cost of capital could shift consumer demand from brown to green products.

Berg, Koelbel, Pavlova, and Rigobon (2022a) demonstrate that this divergence introduces classical measurement error that attenuates estimated effects of ESG performance on stock returns and propose instrumenting ratings from one provider with those of another. My contribution to this literature is to show that even when ESG ratings agree across providers, they can systematically misidentify the environmental impact of the assets a firm finances. In securitization markets, the environmental externality originates from the collateral pool rather than the issuer, a distinction that existing ESG rating methodologies do not capture.

This paper also contributes to the literature on the real effects of captive finance and securitization. Benmelech, Meisenzahl, and Ramcharan (2017) and Ramcharan, Verani, and Van den Heuvel (2016) find that the disruption in ABS markets during the Financial Crisis reduced automobile credit supply and vehicle sales. Klee and Shin (2020) find that lenders signal private information in the auto ABS market by warehousing high-quality loans longer. Benetton et al. (2021) show that vertical integration of manufacturing and credit provision allows manufacturers to increase cash collected from vehicle sales through credit fire sales. Hankins, Momeni, and Sovich (2026) show that captive lending creates a channel for trade policy to affect consumer credit. I measure the pass-through of the ESG convenience yield to consumer rates and explore the impact of ESG investing on consumer loan demand. By examining how differences in the cost of capital for ESG assets influence loan demand, I shed light on whether ESG preferences translate into tangible economic outcomes and what their broader economic implications are.

2 Data

The sample covers all publicly traded consumer loan auto ABS issued from 2017 to 2022, consisting of approximately 17.8 million unique loans from 281 ABS deals of 22 issuers.[13] I exclude vehicle lease and dealer floor plan securitizations due to their different risk characteristics.

ABS Deal Data I collect information about the structure of each deal from prospectuses filed with the SEC, which include details on the deal and its tranches, such as issue date, credit rating, coupon, spreads, issuance amounts, weighted average life (WAL), and book-running banks. I calculate issuance spreads as the difference between the issuance yields and yield curve estimates of Filipović, Pelger, and Ye (2025) by matching the maturity to the WAL. Table 1 presents summary

[13] These issuers are: Ally Financial, AmeriCredit, BMW Financial, Capital One Bank, CarMax, Carvana, Exeter Finance, Fifth Third Bank, Ford Credit, GM Financial, Honda Finance, Hyundai Capital, JM Family (WOART), JM Family (WOSAT), Mechanics Bank, Mercedes-Benz Financial Services, Nissan Finance, Santander Bank (DRIVE), Santander Bank (SDART), Toyota Motor Credit, USAA Federal Savings Bank, and Volkswagen Credit.

Table 1: Summary Statistics of Issuance-Level Data (A-2 Tranches)

	Mean	SD	Median	Min	Max	N
Total Deal Size ($ m)	1,242.38	348.24	1,250.00	367.31	2,663.82	281
Tranche Size ($ m)	366.71	131.99	362.00	42.40	746.94	281
Weighted Average Life / Maturity (Years)	0.98	0.32	1.01	0.37	3.50	281
Issuance Spread (bps)	41.68	29.10	32.29	6.13	194.22	281
Coupon (%)	1.91	1.30	1.86	0.14	5.81	281
Subprime ABS	0.28	0.45	0.00	0.00	1.00	281
Captive Lender	0.44	0.50	0.00	0.00	1.00	281
Number of Receivables in Collateral Pool	63,031.56	21,302.73	62,886.00	15,329.00	136,860.00	281
Loan-to-Value	0.92	0.04	0.92	0.80	0.98	281
Average Credit Score	706.19	74.87	738.34	564.98	788.46	281
Average Interest Rate (%)	7.64	5.87	4.46	1.38	21.35	281
Average Remaining Balance	0.90	0.07	0.91	0.74	1.00	281
Warehousing Time (Months)	9.54	4.38	9.19	1.33	21.06	281
Expected tCO2 per $100,000	292.26	51.02	295.29	161.51	456.01	281
Expected tCO2 per Vehicle	70.23	14.76	67.61	42.94	125.57	281
Financed tCO2 per $100,000	219.14	39.96	211.08	107.10	311.78	281
Financed tCO2 per Vehicle	57.78	12.11	54.50	40.54	101.25	281
Average ESG Score of Issuer	0.58	0.15	0.62	0.15	0.76	243
Average Environmental Score of Issuer	0.62	0.24	0.68	0.01	0.89	243

Notes: This table reports summary statistics for the main variables. The first two columns report the mean and the standard deviation, and the third to fifth columns report the median, minimum, and maximum, respectively. Expected and financed CO_2 emissions are defined in Eq. (1). The sample contains all A-2 tranches of publicly traded consumer loan auto ABS from 2017 to 2022.

statistics for the A-2 tranches of each deal. The average deal size is $1.2 billion, of which the A-2 tranche accounts for 30%. The average spread is 42 bps with a WAL of one year. Captive lenders issue about 44% of deals and approximately 28% are subprime deals. The average deal finances 63,031 vehicles. A $100,000 investment finances 219 tCO_2 over the remaining life of the collateral, on average.

Loan-level Data The loan-level data are from SEC Form ABS-EE. Form ABS-EE is part of the post-financial-crisis reporting requirements under Regulation AB that went into effect in November 2016.[14] This regulation mandates that all issuers of public asset-backed securities must submit loan-level information electronically, with monthly updates on loan pool performance. The data include information on the originator, borrower, and collateral of each loan. Appendix Table A2 presents the summary statistics of the loan-level data. The average borrower in the sample finances $25,822 at 90% loan-to-value, with a 7.84% interest rate and a 67-month term. Their credit score is 708 and their monthly payment-to-income ratio is 0.08. The vehicle the average borrower

[14]Using ABS-EE data, Bena, Bian, and Tang (2023) find that EV loans have higher interest rates, lower loan-to-value ratios, and shorter terms due to higher residual value risk from technological obsolescence. Klee, Morse, and Shin (2023) find that EV borrowers default 29% less than ICE borrowers.

Table 2: Average Securitization Intensity by Industry

Firm-level averages by industry:	Banks	Captive Lenders	Retailers	All Industries
Vehicles securitized per year	278,569	231,265	261,850	248,870
Vehicles securitized as percentage of units sold		16%	39%	20%[1]
Amount securitized as percentage of revenue	83%	32%	24%	45%
Amount securitized as percentage of assets	11%	05%	30%	10%

Notes: This table reports the average securitization intensity by industry for N=60 firm-years. Securitizations include only consumer loans and exclude lease and floor plan securitizations. Revenue and assets are for the U.S. vehicle lending segment when available; otherwise, they are for the overall U.S. segment (Source: S&P Compustat Segment files). Unit sales of manufacturers are from www.goodcarbadcar.net. [1]excludes banks.

is financing is worth $27,341.

CO_2 Emissions Data Data on CO_2 emissions come from the EPA. I match these by make, model, and model year to the loan-level data. Estimates of survival-weighted vehicle miles traveled (SVM) by vehicle type come from the EPA Corporate Average Fuel Economy (CAFE) standard simulator. The average vehicle in the sample is driven 202,834 miles, with 161,660 miles financed. Emissions vary significantly among the collateral, which includes fully electric vehicles, compact cars, SUVs, pickup trucks, and other high-emission vehicles.[15] The average vehicle will emit 62 tCO_2 over its remaining lifetime with a standard deviation of 29.5 tCO_2.

Issuer-level ESG Scores I collect issuer-level ESG scores from MSCI, Sustainalytics, Refinitiv, and Standard and Poor's (S&P).[16] All providers create their scores based on publicly available information and penalize companies with limited reporting. I use the average across available ESG and environmental pillar scores and report results using individual ESG and environmental pillar scores in the Appendix. The scores are available for 17 of the 22 issuers in the sample.

3 Securitized Auto Loans and Their CO_2 Emissions

Auto loan securitizations were among the first consumer ABS to come to market in the 1980s. As of 2022, auto ABS provide capital for approximately 20% of all auto loans in the United States. The auto ABS market is divided into prime and subprime deals based on the creditworthiness of the underlying loans. Issuers of auto ABS come from various industries, including vehicle man-

[15]The 10 most common vehicles in the sample exemplify this heterogeneity. These are, in order, Toyota Camry (sedan, on average 60t of CO_2 emissions over full lifetime), Toyota RAV4 (SUV, 73t), Toyota Corolla (sedan, 53t), Nissan Rogue (SUV, 62t), Chevrolet Silverado (truck, 120t), Honda Civic (sedan, 51t), Nissan Altima (sedan, 59t), Honda CR-V (SUV, 65t), Honda Accord (sedan, 62t), and Ford F-150 (truck, 114t).

[16]Technically, a special purpose vehicle (SPV) issues the auto ABS. The SPVs do not have ESG scores. I use the ESG scores of the sponsor (e.g., Santander) and with a slight abuse of terminology refer to the sponsor as the issuer.

ufacturers and their captive lenders, vehicle retailers, banks, and non-bank finance companies.

Table 2 highlights the importance of consumer loan securitization for these industries. Companies in the sample securitize approximately 45% of their revenues, 10% of their total assets, and 20% of total unit sales annually. Auto loan securitizations are a crucial part of the financial intermediation chain, and changes in financing conditions in the ABS market significantly impact the supply of credit and vehicle sales (Ramcharan et al., 2016, Benmelech et al., 2017).

Compared with corporate and municipal bond markets, the security design of the auto ABS market is highly standardized. Aside from differences in collateral pools, only a few parameters distinguish auto loan securitizations from each other. All 281 deals in the sample are structured to amortize monthly, with higher-seniority tranches receiving repayments first. The high levels of standardization in the auto ABS market and the safety of AAA-rated senior tranches make auto ABS highly liquid. He and Mizrach (2017) report that auto ABS have bid-ask spreads as low as those of agency mortgage-backed securities that trade in the to-be-announced market.[17] Prepayment is the main risk for investors in senior tranches of auto ABS since time and risk tranching, over-collateralization, and other credit enhancements mitigate credit risk. Prepayment risk arises from early loan repayment or borrower defaults leading to vehicle repossession.

Stylized Facts About CO_2 Emissions From Auto ABS The granular loan-level data, which publicly traded auto ABS must disclose, allow me to calculate the financed CO_2 emissions of each collateral pool. The CO_2 emissions financed by ABS b are the sum over the financed emissions of each vehicle i in its collateral pool:

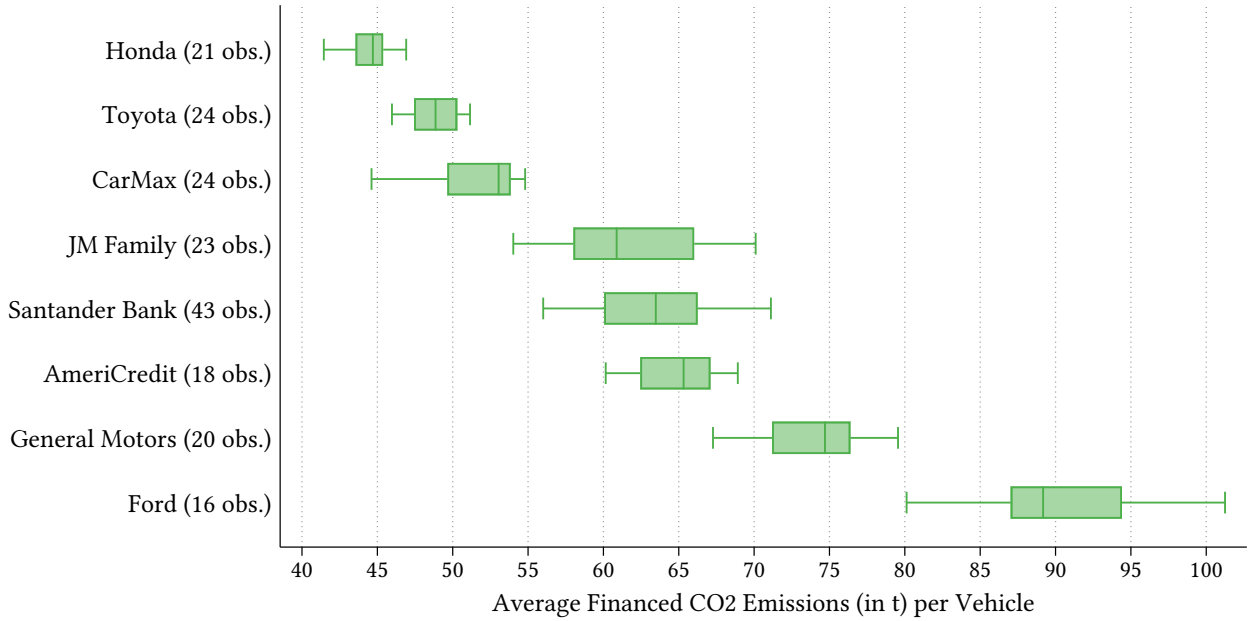
$$\mathbb{E}_t\left[\text{Financed } CO_2 \text{ Emissions}\right]_b = \sum_{i \in b} \underbrace{CO_2 \text{ Emissions per Mile}_i \times \mathbb{E}_t\left[\text{Survival-Weighted Miles}\right]_i}_{\text{Expected Emissions}} \times \underbrace{\text{LTV}_i \times \text{Outstanding Balance Share}_i}_{\text{Financing Adjustment}}. \quad (1)$$

The first term on the right-hand side of Eq. (1) is vehicle i's CO_2 emissions, measured in tons of CO_2 per mile driven. The second term is the expected survival-weighted vehicle miles traveled over the lifetime of the vehicle. The product of these terms is the total expected lifetime emissions of a new vehicle.[18] The loan-to-value (LTV) part of the financing adjustment of Eq. (1) reflects that not all expected CO_2 emissions are financed through a loan since many consumers make down payments at the time of purchase. The financing adjustment also considers that loans have

[17] Online Appendix Figure B1 shows examples of auto ABS deal structures.

[18] I adjust the survival-weighted vehicle miles traveled of used vehicles to reflect the remaining lifetime of the vehicle.

Figure 1: Dispersion of CO_2 Emissions Across all ABS Pools of the Eight Largest Issuers



Notes: This figure presents boxplots of average financed tCO_2 per vehicle for the eight issuers with the largest number of auto ABS deals issued from 2017 to 2022. Together these eight issuers account for 67.3% of the 281 deals in the sample. Issuers are ordered by median financed tCO_2 per vehicle. Financed tCO_2 is defined in Eq. (1); summary statistics across the full sample appear in Table 1. Each observation is one A-2 tranche deal; the box spans the 25th to 75th percentile of deal-level averages for a given issuer, the horizontal line inside the box denotes the median, and whiskers extend to the most extreme deal within 1.5 × IQR. The figure highlights that across-issuer differences in financed emissions are large relative to within-issuer variation, motivating the identification strategy in Section 4.

different outstanding balances at the time of securitization.

Figure 1 highlights that just as motorists choose between high- and low-emission vehicles, investors choose between auto ABS that finance high- and low-emission vehicles. Auto ABS of Honda and Toyota finance less than 50 tCO_2, while those of Ford and General Motors finance more than 74tCO_2 on average. The vehicle type composition of the collateral pool explains the large differences in emissions. Appendix Table A1 shows that a 1 percentage point increase in the share of trucks in the collateral pool raises the average CO_2 per vehicle by 1.02 tons.

Table 3 shows that ESG scores, commonly used by the asset management industry to assess the environmental impact of investments, correlate positively with CO_2 emissions and other measures of environmental impact. The positive correlation between ESG scores and CO_2 emissions creates problems if investors use ESG scores to screen green from brown auto ABS. The finding that ESG scores correlate positively with CO_2 emissions complements Berg et al. (2022b), who document substantial divergence across ESG rating providers. While their work shows that ratings disagree about how green a firm is, my results show that even when ratings agree,

Table 3: Pairwise Correlations for Measures of Greenness: ESG, CO_2, and Miles-per-Gallon

	Average ESG Score	Average Env. Score	Fin. tCO2 per USD	Fin. tCO2 per Vehicle	Truck Share	MPG ×(-1)	EPA GHG Score ×(-1)
Average ESG Score	1.00						
Average Env. Score	0.94***	1.00					
Fin. tCO2 per USD	0.13**	-0.04	1.00				
Fin. tCO2 per Vehicle	0.20***	0.19***	0.18***	1.00			
Share of Trucks	0.05	-0.02	0.29***	0.84***	1.00		
MPG×(-1)	0.12*	-0.02	0.40***	0.68***	0.74***	1.00	
EPA GHG Score×(-1)	0.01	-0.06	0.09	0.78***	0.86***	0.82***	1.00

Notes: This table reports pairwise Pearson correlation coefficients between the average ESG scores, average environmental pillar scores, financed CO_2, share of trucks in collateral pool, negative of MPG, and negative of EPA GHG score. Average ESG and environmental scores are averages of MSCI, Sustainalytics, S&P, and Refinitiv. * p<0.10, ** p<0.05, *** p<0.01.

they can misidentify the environmental impact of the assets a firm finances. This distinction between entity-level and asset-level greenness is particularly consequential in securitization markets, where the environmental externality originates from the collateral rather than the issuer.

4 Issuance Spreads, ESG Basis, and ESG Convenience Yield

4.1 A Stylized Green Asset Pricing Model

I build a stylized asset pricing model with an ESG convenience yield in the spirit of Krishnamurthy and Vissing-Jorgensen (2012) and derive the difference in yields between green and brown assets. The economy is populated by a single investor whose Euler equation is

$$\mathbb{E}_t \left[M_{t+1} R^i_{t+1} \right] = \exp(-\beta^i \lambda_t). \tag{2}$$

The expression on the left side of the equation is standard. On the right side, I allow the investor to derive a convenience yield $\lambda_t \geq 0$ from holding asset i with greenness $\beta^i \in [0, 1]$. Assets with higher values of β^i are greener and earn a convenience yield of $\beta^i \lambda_t$. The convenience yield is asset-specific and hence cannot be folded into the stochastic discount factor M_t. For simplicity, I assume that there are only two assets in the economy: a brown asset b and a green asset g with $\beta^g > \beta^b$. I assume that $m_t = \log M_t$ and $r^i_t = \log R^i_t$ are conditionally normal. Rewriting the Euler equation using log-normality, one finds

$$\mathbb{E}_t \left[m_{t+1} \right] + \frac{1}{2} \operatorname{Var}_t \left[m_{t+1} \right] + \mathbb{E}_t \left[r^i_{t+1} \right] + \frac{1}{2} \operatorname{Var}_t \left[r^i_{t+1} \right] + \operatorname{Cov}_t \left[m_{t+1}, r^i_{t+1} \right] + \beta^i \lambda_t = 0$$

and the following result:

Lemma 1. *The expected return on a long position is decreasing in the convenience yield and in the greenness of the asset:*

$$\mathbb{E}_t\left[r^i_{t+1}\right] - r^f_t + \sigma^2_{i,t}/2 = -\sigma_{i,m,t} - \beta^i \lambda_t \tag{3}$$

Using the Campbell and Shiller (1988) approximation, one can write the dividend yield of an asset with fixed maturity T as

$$\begin{aligned} dp^i_t &\approx \sum_{j=0}^{T-1} \rho^j \mathbb{E}_t\left[r^i_{t+1+j}\right] - \sum_{j=0}^{T-1} \rho^j \mathbb{E}_t\left[\Delta d^i_{t+1+j}\right] - \kappa \frac{1-\rho^T}{1-\rho} + \rho^T \mathbb{E}_t\left[dp^i_{t+T}\right] \\ &= -\beta^i \lambda_t \frac{1-\rho^T}{1-\rho} + \sum_{j=0}^{T-1} \rho^j r^f_{t+j} - \sum_{j=0}^{T-1} \rho^j \left(\sigma_{i,m,t,t+j} + \sigma^2_{i,t,t+j}/2\right) - \sum_{j=0}^{T-1} \rho^j \mathbb{E}_t\left[\Delta d^i_{t+1+j}\right] - \kappa \frac{1-\rho^T}{1-\rho}, \end{aligned} \tag{4}$$

with $\rho = \frac{1}{1+\exp\left(\overline{d-p}\right)}$ and $\kappa = -\log(\rho) - (1-\rho)\log(1/\rho - 1)$. The first term in Eq. (4) shows that a higher non-pecuniary value derived from the greenness of an asset lowers the dividend yield and raises the price of the asset. Taking the difference of Eq. (4) between a green and a brown asset with identical payoffs and risk, one finds the ESG basis spread:

Lemma 2. *Suppose the green and brown assets have identical expected cash flows and risk exposures. The difference in their dividend-price ratios defines the ESG basis spread,*

$$dp^g_t - dp^b_t = -(\beta^g - \beta^b)\,\lambda_t\,\frac{1-\rho^T}{1-\rho}, \tag{5}$$

whose absolute value is increasing in the final maturity T, the convenience yield λ_t, and the greenness gap $\beta^g - \beta^b$.

In the case of a one-period bond, the ESG basis spread simplifies to:

$$dp^g_t - dp^b_t = -(\beta^g - \beta^b)\lambda_t$$

Eq. (4) shows that to infer the ESG basis spread, one needs to carefully account for differences in risk exposure and cash-flow growth of green and brown assets. Below, I build an identification strategy that exploits the features of the auto ABS market to isolate the ESG basis.

4.2 Identification Strategy for ESG Basis Spread

My identification strategy for the ESG basis rests on three points. First, high standardization and the short-term, safe-asset nature of the securities minimize the risk that unobserved heterogeneity affects the estimates. Second, the seniority structure and design of securitizations ensure that prepayment is the main risk factor for senior tranches. Allocating cash flows across tranches and time shifts credit risks to subordinate tranches. My analysis focuses on senior tranches rated AAA by at least two agencies. Credit losses would need to reach about 50%, assuming zero recovery value, to affect these tranches.[19] Third, borrower and loan characteristics determine prepayment risk, not the greenness of the issuer or the collateral. Borrowers with high-interest-rate loans prepay when interest rates fall, regardless of the ESG score of the loan originator or the CO_2 emissions of the collateral they finance. Moreover, the granularity of the loan-level data allows me to use ex-post prepayments as placebo controls. This alleviates concerns that investors use greenness to infer risky payoffs, as in Pedersen et al. (2021), rather than to express their non-pecuniary preferences.[20]

Empirical Specifications I estimate the ESG basis spread using the following specification:

$$\log\left(\text{Issuance Spread}\right)_b = \varphi \log\left(\text{Greenness}\right)_b + \mathbf{X}_b'\zeta + \gamma_{p,t} + \varepsilon_b \tag{6}$$

for ABS b issued in year-month t.[21] The coefficient of interest, φ, is the elasticity of issuance spreads with respect to greenness. The (sub)prime-by-year-month fixed effect, $\gamma_{p,t}$, identifies φ using variation in greenness across (sub)prime securitizations issued during the same month.

The main control variables in $\mathbf{X}_b$ are known predictors of prepayment risk. The predictors are collateral pool averages of LTV, credit scores, remaining loan balance, interest rate, and vehicle values. I further include the average time that loans have been warehoused on the lender's balance sheet before securitization, as lenders often warehouse higher-quality loans for longer to signal private information to investors (Klee and Shin, 2020). To control for within-month variation in market conditions, I include the six-month yield from Filipović et al. (2025), the level of the VIX on the day of issuance, and the standard deviation of the VIX in the 30 days before issuance in

[19] If a borrower defaults, the vehicle is repossessed and sold. For senior tranches, this process is an "involuntary prepayment." The most junior tranche bears the difference between the outstanding balance and the recovery value, with historical recovery values around 60% for prime and 45% for subprime loans (Structured Finance Association).

[20] Ex-post prepayment and delinquency measures are included as placebo controls for robustness. These outcomes were not known to investors at issuance and are not used to interpret primary market pricing. Their inclusion serves solely to verify that the ESG coefficient is not capturing unobserved differences in realized performance.

[21] I find consistent results using a specification with an indicator variable equal to one if the greenness is above the 50th percentile. See Online Appendix Table B2 for OLS and Table B3 for a propensity score estimator.

Table 4: Ex-post Performance of Collateral Pools, ESG scores, and CO_2 Emissions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	|Δ Realized to Assumed Prepayment|				Realized % loans delinquent 30d+			
Financed tCO2 per $100,000	0.028				0.026			
	(0.086)				(0.029)			
Financed tCO2 per Vehicle		-0.043				-0.022		
		(0.126)				(0.015)		
ESG Score			0.075				0.071	
			(0.095)				(0.064)	
Environmental Score				-0.015				0.087
				(0.079)				(0.079)
Year-month FE	✓	✓	✓	✓	✓	✓	✓	✓
Subprime FE					✓	✓	✓	✓
Adj. R^2	0.20	0.20	0.23	0.23	0.97	0.98	0.98	0.98
Observations	281	281	243	243	281	281	243	243

Notes: This table reports results from a test of the identifying assumption that measures of greenness are uncorrelated with risk factors. The outcome variable in columns (1) to (4) is the absolute difference between the realized prepayment speed and the assumed prepayment speed. The outcome variable in columns (5) to (9) is the realized 30+ days delinquency rate as a proxy for involuntary prepayment through default. Columns (5) to (9) control for the average credit score of the collateral pool and subprime fixed effects. ESG and environmental pillar scores are averages over scores from MSCI, Sustainalytics, Refinitiv, and S&P. Coefficients are standardized to unit variances. Standard errors in parentheses are clustered at year-month and originator. * p<0.10, ** p<0.05, *** p<0.01.

$\mathbf{X}_b$. All variables are in natural logarithms. The specifications include assumed absolute prepayment speed (APS) fixed effects, interacted with the tranche's weighted average life to allow for potential "ramp-up" periods in which prepayments increase before leveling off at their assumed APS. Online Appendix Table B4 shows that using the Lasso estimator of Belloni, Chernozhukov, and Hansen (2014) to select from over 850 potential control variables yields similar results.

Identifying Assumption The identifying assumption in Eq. (6) is that the assignment of greenness is uncorrelated with the error term conditional on risk factors: once risk and security design are accounted for, the assignment of greenness is "as good as random." This identification assumption allows me to infer the ESG basis spread from variation in greenness across securitizations. The null hypothesis is that greenness does not affect issuance spreads, implying φ=0. Evidence that φ<0 indicates that investors accept lower yields because they prefer greener assets.

Auto ABS issuers release monthly performance reports after issuance that allow me to test whether greenness correlates with ex-post performance. Specifically, I examine two measures that capture voluntary and involuntary prepayment at the collateral-pool level: the absolute realized

difference in monthly prepayment speed compared with its prospectus assumption, and the average realized percentage of loans more than 30 days delinquent. Table 4 shows that neither CO_2 emissions nor ESG scores predict the performance of collateral pools. The estimates are noisy and close to zero. Although ultimately untestable, these results support the identification assumption: measures of greenness are not informative about the ex-post performance of auto ABS.

4.3 Results

Table 5 presents the results of the pricing model of Eq. (6). Odd-numbered columns control for predictors of prepayment risk only, even-numbered columns add controls for ex-post realizations of prepayment risk. Panel A shows estimates of the elasticity of issuance spreads with respect to either ESG scores or CO_2 emissions using the pricing model of Eq. (6). Panel B runs a horse race between CO_2 emissions and ESG scores, comparing their effects on issuance spreads.

Panel A of Table 5 shows that both high ESG scores and high CO_2 emissions predict lower issuance spreads. The spreads have an elasticity of -0.33 to -0.49 for ESG scores and -0.16 to -0.2 for CO_2 emissions intensity. I find similar results using individual ESG scores from MSCI, Sustainalytics, Refinitiv, and S&P instead of the average ESG score. Appendix Table A3 shows that all ESG and environmental pillar scores are associated with lower issuance spreads. Panel B presents the results of a horse race between CO_2 and ESG in pricing auto ABS. The elasticity with respect to ESG scores remains stable and significant, but the elasticity of CO_2 emissions shrinks toward zero and loses statistical significance. This suggests that investors rely on ESG scores to identify and price green assets. However, CO_2 emissions and ESG scores correlate positively. Consequently, investors who rely on ESG scores to allocate capital inadvertently subsidize CO_2 emissions, as Panel A shows. Reassuringly, across both panels, estimates accounting for ex-post performance of collateral pools are similar to those controlling only for predictors of prepayment risk. This further supports the assumption that neither ESG scores nor CO_2 emissions predict the performance of auto ABS.

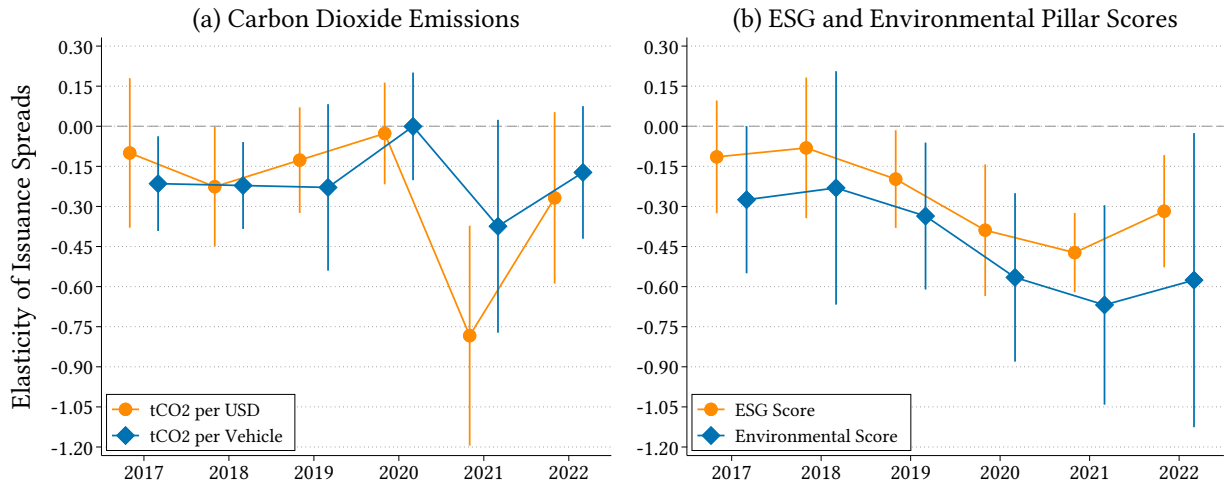
Figure 2 shows elasticity estimates of issuance spreads with respect to CO_2 emissions, ESG scores, and environmental pillar scores by year. The CO_2 elasticities in Panel (a) are mostly negative, but they are noisy and do not display a clear monotonic time trend. By contrast, Panel (b) shows a clearer pattern: the elasticities with respect to ESG scores and, even more so, environmental pillar scores become more negative from 2017 to 2021, before moderating somewhat in 2022. Throughout, elasticity estimates for environmental pillar scores tend to be more negative than those for composite ESG scores, consistent with the hypothesis that investors prioritize "environmental"

Table 5: The Pricing of Greenness in Auto Loan Securitizations

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
				Issuance Spread				
Panel A: Elasticity of Issuance Spreads With Respect to Either ESG Score or Carbon Emissions								
ESG Score	-0.337***	-0.380***						
	(0.104)	(0.101)						
Environmental Score			-0.411**	-0.483***				
			(0.165)	(0.165)				
Financed tCO2 per $100,000					-0.187*	-0.195**		
					(0.095)	(0.093)		
Financed tCO2 per Vehicle							-0.158*	-0.169*
							(0.085)	(0.082)
Year-month FE, daily market controls	✓	✓	✓	✓	✓	✓	✓	✓
Prepayment speed FE, tranche controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-ante prepayment controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-post prepayment controls		✓		✓		✓		✓
Adj. R^2	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98
Observations	243	243	243	243	281	281	281	281
Panel B: Elasticity of Issuance Spreads With Respect to ESG Score and Carbon Emissions								
ESG Score	-0.337***	-0.376***	-0.331***	-0.371***				
	(0.098)	(0.095)	(0.101)	(0.099)				
Environmental Score					-0.417**	-0.481***	-0.402**	-0.467**
					(0.165)	(0.163)	(0.164)	(0.164)
Financed tCO2 per $100,000	-0.169	-0.157			-0.173	-0.165		
	(0.105)	(0.111)			(0.101)	(0.107)		
Financed tCO2 per Vehicle			-0.127	-0.115			-0.134	-0.125
			(0.092)	(0.097)			(0.089)	(0.094)
Year-month FE, daily market controls	✓	✓	✓	✓	✓	✓	✓	✓
Prepayment speed FE, tranche controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-ante prepayment controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-post prepayment controls		✓		✓		✓		✓
Adj. R^2	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98
Observations	243	243	243	243	243	243	243	243

Notes: This table reports estimates of the effects of greenness on issuance spreads of auto ABS. All variables are in logs. ESG and environmental pillar scores are averages over scores from MSCI, Sustainalytics, Refinitiv, and S&P. Panels A and B show coefficient estimates of Eq. (6). Daily market controls include 6-month yield from Filipović et al. (2025), the level of the VIX on the issuance date, and the standard deviation of the VIX in the 30 days before the issuance date. Ex-ante prepayment controls are collateral pool averages of LTV, credit scores, remaining loan balance share, interest rate, and vehicle values. Ex-post prepayment controls are the difference between realized and assumed prepayment and the realized 30+ days delinquency rate. Standard errors in parentheses are clustered at year-month and originator. * p<0.10, ** p<0.05, *** p<0.01.

Figure 2: Elasticity with Respect to CO_2, ESG, and Environmental Score From 2017 to 2022



Notes: This figure shows yearly elasticity estimates and 90% confidence intervals of the risk-adj. model of Eq. (6). Panel (a) shows elasticity estimates for CO_2 per vehicle and CO_2 per USD. Panel (b) shows elasticity estimates for the ESG and environmental scores. ESG and environmental pillar scores are averages over scores from MSCI, Sustainalytics, Refinitiv, and S&P.

impact, although the confidence intervals overlap substantially.

Translating the ESG Basis Spread into an ESG Convenience Yield The estimated differences in issuance spreads induced by high ESG scores (i.e., the ESG basis spread) translate into a convenience yield that an investor earns on their ESG investment. Rearranging Eq. (5), one finds that the ESG convenience yield is given by

$$\lambda_t = -\frac{y_t^g - y_t^b}{\beta_t^g - \beta_t^b} \tag{7}$$

in which $y^g - y^b$ is the ESG basis spread and $\beta_t^g - \beta_t^b$ is the difference in ESG scores.

Table 6 shows estimates of the ESG convenience yield over time. The ESG convenience yield is close to zero and not statistically significant at the beginning of the sample. Starting in 2020, the ESG convenience yield becomes statistically significant. The average ESG convenience yield is 0.39% p.a. over the sample period. Consistent with this estimate, Avramov et al. (2024) find an ESG convenience yield for stocks between 0.37% and 0.66%.

4.4 What Drives the Pricing of ESG Scores in the Auto ABS market?

Media Attention to Climate Change I interact ESG and environmental scores with the Media Climate Change Concerns (MCCC) Index of Ardia et al. (2023) to test whether concerns about

Table 6: Estimates of the ESG Convenience Yield From 2017 to 2022

		2017	2018	2019	2020	2021	2022	Avg.
Difference in Average ESG Score:	$\beta_t^g - \beta_t^b$	0.25	0.20	0.22	0.22	0.35	0.23	0.23
ESG Basis Spread (in bps):	$y_t^g - y_t^b$	-2.1	-0.4	-3.8*	-9.6***	-10.2***	-11.2***	-9.1***
		(3.5)	(2.2)	(2.0)	(3.6)	(2.2)	(3.5)	(2.5)
ESG Convenience Yield (in bps):	λ_t	8.7	2.3	17.3*	43.7***	28.8***	49.5***	39.4***
		(14.4)	(11.0)	(8.8)	(16.6)	(6.3)	(15.6)	(11.1)

Notes: This table reports estimates of the ESG convenience yield from 2017 to 2022. Differences in ESG scores and the ESG basis spread are evaluated at the 20th and 80th percentiles of average ESG scores. * p<0.10, ** p<0.05, *** p<0.01.

climate change are associated with the pricing of ESG scores. The indices are constructed using news about climate change published by major U.S. newspapers and newswires. The data include sub-indices that capture specific dimensions, such as societal concerns, environmental impact, and industry-specific transition risks. Ardia et al. (2023) find that increases in climate change concerns are associated with an increase (decrease) in the discount rate of brown (green) stocks.

Table 7 shows that concerns about climate change correlate with the pricing of the ESG convenience yield into auto ABS issuance spreads. I find evidence that media attention to climate change and its environmental consequences significantly influences ESG score pricing. These results lend further support to the hypothesis that the pricing of non-pecuniary preferences is driven by concerns surrounding the environmental impact of climate change. Importantly, media attention to transition risk does not correlate with the effects of ESG and environmental scores on issuance spreads, as shown in Appendix Table A4.

Capital Flows Into ESG Funds van der Beck (2026) reports that more than $1.1 trillion flowed into ESG funds over the past decade.[22] I test whether capital flows into ESG funds correlate with the relative pricing of ESG scores by interacting ESG scores with cumulative net flows into ESG funds since 2012 from van der Beck (2026). Table 7 confirms that they do: a $100bn increase in cumulative net flows is associated with a 2 bps reduction in issuance spreads (columns 1 and 2) for issuers with relatively higher ESG scores.

Portfolio Holdings of ESG Mutual Funds I analyze the portfolios of ESG mutual funds to directly test whether the greenness of auto ABS influences their investment decisions in Appendix A.4. The portfolio data show that ESG funds invest more in auto ABS from issuers with high ESG scores than non-ESG funds. While this is not surprising by itself, the positive corre-

[22]Even mutual funds without declared ESG objectives are affected by marketwide ESG concerns: Hartzmark and Sussman (2019) use the introduction of Morningstar's sustainability ratings for mutual funds to show that being categorized by a third party as either high or low sustainability results in large net capital flows.

Table 7: The Effects of ESG Capital Flows and Climate Change Concerns on Spreads of Auto ABS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
				Issuance Spread				
ESG Score	-0.222*	-0.227*	-0.116	-0.125				
	(0.125)	(0.127)	(0.161)	(0.161)				
ESG Capital Flow ($100bn) × ESG Score	-0.033***	-0.034***						
	(0.011)	(0.011)						
Env. Concerns News Index × ESG Score			-0.322*	-0.392**				
			(0.180)	(0.178)				
Environmental Score					-0.616***	-0.620***	-0.215	-0.283
					(0.179)	(0.185)	(0.229)	(0.234)
ESG Capital Flow ($100bn) × Env. Score					-0.029*	-0.030*		
					(0.015)	(0.014)		
Env. Concerns News Index × Env. Score							-0.398*	-0.485**
							(0.213)	(0.222)
Year-month FE, daily market controls	✓	✓	✓	✓	✓	✓	✓	✓
Prepayment speed FE, tranche controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-ante prepayment controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-post prepayment controls		✓		✓		✓		✓
Adj. R^2	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98
Observations	201	201	243	243	201	201	243	243

Notes: This table reports elasticity estimates with respect to ESG scores interacted with flows into ESG funds. ESG and environmental pillar scores are averages over scores from MSCI, Sustainalytics, Refinitiv, and S&P. ESG capital flows are measured in units of $100bn. Flows into ESG funds from 2012 to 2021 are from van der Beck (2026). The Environmental Concerns News Index is a sub-index of the Media Climate Change Concerns News Index of Ardia et al. (2023). Standard errors in parentheses are clustered at year-month. * p<0.10, ** p<0.05, *** p<0.01.

lation between ESG scores and CO_2 emissions of the collateral means that even dedicated ESG funds inadvertently invest more in high-emission auto ABS compared with non-ESG funds. I document that ESG mutual funds hold positions across the full distribution of CO_2 emissions and invest more in high-emission deals relative to non-ESG funds. However, similar to the findings in Table 5, the positive correlation between ESG scores and CO_2 emissions confounds these estimates. Once I control for ESG scores, CO_2 emissions do not have explanatory power for the relative portfolio holdings of ESG mutual funds.

4.5 The ESG Basis Spread in Deals Intermediated by SBTi Banks

Auto ABS are a heavily intermediated asset class. A small group of lead underwriters, drawn from the largest global banks, markets each deal, collects and assesses investor orders, advises issuers on pricing and tranche placement, and coordinates final allocations. Through this bookbuilding role, lead underwriters translate investor demand into the final terms of each securitization.

Lead underwriters may affect the ESG basis spread through two channels. First, they retain

discretion over pricing and allocation (Siani, 2025), so an underwriter with its own climate commitment may use that discretion in ways that favor high-ESG issuers. Second, climate-committed underwriters may have stronger relationships with ESG-sensitive investors and may therefore be better able to generate demand for securities issued by firms with stronger ESG profiles. Both channels imply that the ESG basis spread should be larger in deals intermediated by a climate-committed underwriter, especially when ESG preferences are more salient in financial markets.

To test this mechanism, I classify an underwriting bank as climate-committed if it has made a Science Based Targets initiative (SBTi) commitment.[23] I define SBTi Underwriter$_b$ as an indicator equal to one if at least one lead underwriter of deal b has an SBTi commitment at the time of issuance. In the sample, 40% of deals have at least one SBTi underwriter. Identification comes from the fact that issuers often rotate across lead underwriters, including within the same calendar year.[24] This within-issuer, within-year variation allows me to isolate the effect of underwriter climate commitments from issuer ESG scores, collateral composition, and market conditions. I extend the model of Eq. (6) by interacting ESG scores with the SBTi Underwriter indicator:

$$\log\left(\text{Issuance Spread}\right)_b = \varphi_0\,\text{SBTi}_b + \varphi_1\,\log\left(\text{Greenness}\right)_b + \varphi_2\,\text{SBTi}_b \times \log\left(\text{Greenness}\right)_b + \mathbf{X}_b'\zeta + \gamma_{p,t} + \varepsilon_b. \tag{8}$$

The coefficient φ_2 captures the incremental elasticity in deals placed by SBTi-committed underwriters. I further allow φ_2 to vary with the Environmental Concerns Index of Ardia et al. (2023).

Table 8 reports the results. Columns (1) and (2) show that the ESG basis spread is 60% larger in deals placed by at least one SBTi-committed underwriter. The elasticity of issuance spreads with respect to ESG scores is −0.39 to −0.36 in deals without an SBTi underwriter and an additional −0.23 in deals with one, so that the total elasticity reaches approximately −0.62 to −0.59. Columns (5) and (6) find a comparable pattern for environmental pillar scores. The main effect of the SBTi indicator is statistically zero, indicating that SBTi underwriters do not price deals differently on average. Columns (3), (4), (7), and (8) add a triple interaction with the Environmental Concerns News Index. The incremental SBTi-ESG elasticity becomes more negative when news about the environmental impact of climate change is elevated. The pattern matches the results

[23] The Science Based Targets initiative (SBTi) validates corporate emissions-reduction targets as consistent with the Paris Agreement. For financial institutions, SBTi-validated commitments cover both operating emissions and the financed emissions of lending and investment portfolios.

[24] An example is Ford's 2022 auto ABS issuance program. FORD 2022-A was led by Citigroup, Credit Agricole, and SMBC Nikko; FORD 2022-B by BNP Paribas, RBC Capital Markets, and Societe Generale; FORD 2022-C by SMBC Nikko, Barclays, and Lloyds; and FORD 2022-D by J.P. Morgan, Citigroup, and Scotiabank. Underlined banks had an SBTi commitment at the time of issuance.

Table 8: The ESG Basis Spread in Deals Intermediated by SBTi Banks

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Issuance Spread							
ESG Score	-0.390**	-0.362*	-0.320*	-0.302*				
	(0.166)	(0.171)	(0.155)	(0.164)				
Env. Score					-0.589***	-0.470**	-0.536***	-0.459**
					(0.165)	(0.195)	(0.155)	(0.181)
SBTi Underwriter	-0.025	-0.021	-0.029	-0.026	-0.005	-0.003	-0.002	-0.000
	(0.028)	(0.026)	(0.025)	(0.023)	(0.030)	(0.029)	(0.021)	(0.020)
SBTi × ESG Score	-0.231*	-0.231*	-0.217*	-0.220*				
	(0.129)	(0.132)	(0.110)	(0.114)				
SBTi × Env. Score					-0.437**	-0.417*	-0.458***	-0.456***
					(0.199)	(0.197)	(0.130)	(0.129)
SBTi × Env. Concerns Index			0.012	0.008			0.075	0.073
			(0.053)	(0.055)			(0.054)	(0.054)
SBTi × ESG Score × Env. Concerns Index			-0.598*	-0.541*				
			(0.283)	(0.302)				
SBTi × Env. Score × Env. Concerns Index							-1.254**	-1.206**
							(0.434)	(0.434)
Year-month FE, daily market controls	✓	✓	✓	✓	✓	✓	✓	✓
Prepayment speed FE, tranche controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-ante prepayment controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-post prepayment controls		✓		✓		✓		✓
Adj. R^2	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97
Observations	243	243	243	243	243	243	243	243

Notes: This table reports estimates of Eq. (8). The dependent variable is the log issuance spread of A-2 tranches. SBTi Underwriter is an indicator equal to one if at least one lead underwriter of the deal has an SBTi commitment at the time of issuance. ESG and environmental pillar scores are averages over scores from MSCI, Sustainalytics, Refinitiv, and S&P, in logs and demeaned. The Environmental Concerns News Index is the sub-index of Ardia et al. (2023) capturing news about the environmental impact of climate change, demeaned. All specifications include year-month, subprime, and assumed prepayment speed fixed effects; tranche controls; daily market controls (6-month yield, VIX level, 30-day VIX volatility); and ex-ante prepayment controls (collateral-pool averages of LTV, credit score, remaining balance, interest rate, and vehicle value). Even-numbered columns additionally control for ex-post realized prepayment deviation and 30+ day delinquency. Standard errors in parentheses are clustered at year-month and originator. * p<0.10, ** p<0.05, *** p<0.01.

in Table 7 and suggests that SBTi underwriters amplify the pricing of ESG preferences in periods when those preferences are most salient.

SBTi banks capture part of the surplus through higher underwriting discounts. Appendix Table A7 estimates the effect of SBTi-committed underwriters on the underwriting discount. SBTi-committed underwriters charge 5% to 9% higher underwriting discounts than their peers on otherwise comparable deals, corresponding to 1 to 2 additional basis points per deal. The fee premium rises with ESG demand: a $100bn increase in cumulative ESG flows (van der Beck, 2026) raises the SBTi underwriter premium by an additional 1.0%, and a one standard deviation (≈0.28) increase in either the aggregate or environmental concerns news index of Ardia et al. (2023) adds

3.8%. Consistent with a model in which SBTi Banks act as gatekeepers to ESG-sensitive investor demand, their fees rise when ESG demand is stronger.

4.6 Does ESG Investing Intend to Raise the Cost of Emitting CO_2?

The results above imply that ESG investors use ESG scores to express their non-pecuniary preferences over greenness. However, relying on ESG scores to proxy for CO_2 emissions inadvertently subsidizes high-emission securitizations, because ESG scores and collateral-pool emissions correlate positively. This misallocation need not reflect investors' intentions. ESG investors may well intend to raise the cost of emitting CO_2 but lack real-time information about the emissions content of the collateral they finance.

I develop a stylized model of subjective (and potentially biased) beliefs of ESG investors and their intention to price CO_2 emissions in auto ABS. I assume that the CO_2 emissions of auto ABS collateral pools are unobservable to investors. Instead, investors rely on ESG scores to proxy for CO_2 emissions and form subjective expectations $\mathbb{E}^\star[\cdot]$ about the relationship between CO_2 emissions and ESG scores as $\mathbb{E}^\star[\text{CO}_2 \mid \text{ESG Score}] = \varphi^\star \times$ ESG Score with $\varphi^\star < 0$, i.e., ESG investors expect high ESG scores to be associated with low CO_2 emissions. Using ESG scores as proxy variables for CO_2 emissions, the expected effect of CO_2 emissions on issuance spreads under the subjective expectations of ESG investors is

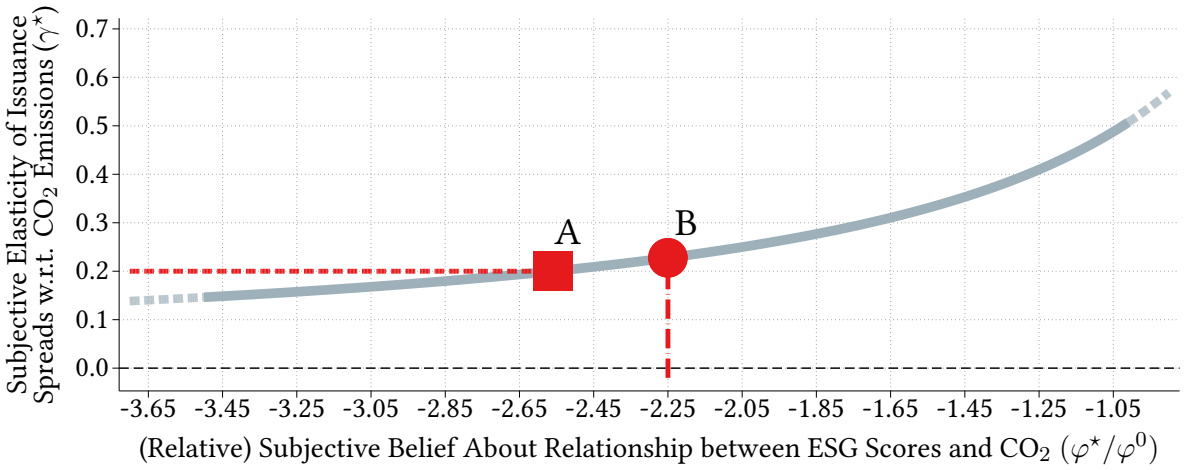
$$\begin{aligned}\mathbb{E}^\star[\text{Issuance Spread} \mid \text{ESG Score}] &= \gamma^\star \times \mathbb{E}^\star[\text{CO}_2 \mid \text{ESG Score}] \\ &= \gamma^\star \times \varphi^\star \times \text{ ESG Score} \\ &= \alpha^\star \times \text{ ESG Score}\end{aligned}$$

in which $\gamma^\star > 0$ and $\alpha^\star < 0$ reflect ESG investors' non-pecuniary preferences. Let φ^0 be the true relationship between CO_2 and ESG scores observed by the econometrician and define δ as

$$\delta \equiv \alpha^\star/\varphi^0 = \underbrace{\gamma^\star}_{\text{Intended Effect}} \times \underbrace{\varphi^\star/\varphi^0}_{\text{Subjective Beliefs}} . \tag{9}$$

This δ can be estimated with an instrumental variable strategy by using ESG scores to instrument for CO_2 emissions. Note that the instrumental variable estimate is not the causal effect of actual CO_2 emissions on issuance spreads. Instead, the estimate reflects the subjective intended effect of CO_2 emissions, $\gamma^\star$, scaled by the subjective efficacy of ESG scores, $\varphi^\star$, relative to the true relation-

Figure 3: Curve of Intended Effect ($\gamma^\star$) as a Function of Subjective Beliefs ($\varphi^\star/\varphi^0$)



Notes: This figure shows in grey the curve of $\gamma^\star = \hat{\delta} \times \left(\varphi^\star/\varphi^0\right)^{-1}$ in which $\hat{\delta} \approx -0.51$ is from Column (5) of Appendix Table A5. Point A marks $\gamma^\star_A \approx 0.2$ implied by the subjective elasticity of wealth with respect to CO_2 emissions of investors surveyed in Haber et al. (2022). Point B marks $\varphi^\star_B/\varphi^0 \approx -2.25$ implied by a representativeness heuristic (Gennaioli and Shleifer, 2010) that uses the average elasticity coefficients of firm-level CO_2 emissions with respect to firm-level ESG scores from Appendix Table A6. See section 4.6 for details.

ship observed by the econometrician, φ^0.[25] Eq. (9) makes clear that the magnitude of δ is shaped by two forces: the extent to which investors intend to price CO_2 emissions into issuance spreads, summarized by $\gamma^\star$, and their subjective beliefs about how well ESG scores identify greener securities, summarized by $\varphi^\star/\varphi^0$. A large value of δ therefore does not, by itself, imply a strong preference for lower emissions. It may instead reflect that investors treat ESG scores as a more informative proxy for underlying emissions than the relationship in the data would support.[26]

I recover $\hat{\delta}$ by instrumenting CO_2 emissions with ESG scores using the specification of Eq. (6). Appendix Table A5 shows that when investors use ESG scores to price the CO_2 emissions of auto loan securitization, it results in a lower cost of capital for high-emission securitizations. The $\hat{\delta}$ coefficients are negative and statistically significant: a 1% higher CO_2 emissions intensity is associated with approximately 0.51% lower issuance spreads.

Figure 3 plots the curve implied by $\hat{\delta} \approx -0.51$, which maps ESG investors' subjective beliefs about the relationship between ESG scores and CO_2 emissions into the intended effect of emissions on issuance spreads. The mapping also runs in reverse: for any intended effect of CO_2 emissions on spreads, one can infer the corresponding subjective beliefs about the ESG-emissions relationship.

[25] Formally, the data generating process is: Issuance Spread $= \beta \times CO_2 + \alpha^\star \times \text{ESG} + \varepsilon$ with β=0 and $\alpha^\star$<0. Instrumenting CO_2 emissions with ESG scores in the model converges in probability to $\delta \equiv \beta^{IV} \xrightarrow{p} \beta + \alpha^\star/\varphi^0$ where φ^0 is the first-stage regression coefficient of $CO_2 = \varphi^0 \times \text{ESG} + \nu$.

[26] Assuming $\gamma^\star > 0$, the sign of δ depends on investors' subjective beliefs about the relationship between ESG scores and CO_2 emissions.

I locate ESG investors on the curve using two complementary approaches. First, I rely on the elasticity of wealth with respect to CO_2 emissions reported by investors surveyed in Haber et al. (2022) to infer the intended pricing of CO_2 emissions into issuance spreads: 52% of investors are willing to give up 10% of their wealth to have the companies they are invested in change from industry-standard carbon emissions levels to "net zero" by 2050. That is, the majority of surveyed investors are willing to give up 0.2% of their wealth to reduce CO_2 emissions by 1%.[27] I thus set $\gamma_A^\star \approx 0.2$, highlighted in Point A in Figure 3, which implies $\varphi^\star/\varphi^0 \approx -2.6$ under $\hat{\delta} \approx -0.51$. Second, I estimate investors' subjective beliefs about the relationship between ESG scores and CO_2 emissions, $\varphi^\star$, and use these beliefs to back out the implied $\gamma^\star$. A natural benchmark is that ESG investors form $\mathbb{E}^\star[CO_2 \mid \text{ESG Score}]$ using a representativeness heuristic (Gennaioli and Shleifer, 2010), extrapolating from the firm-level relationship between CO_2 emissions and ESG scores to auto ABS. This suggests approximating $\varphi^\star$ with the elasticity of firm-level CO_2 emissions intensity with respect to ESG and environmental scores. Appendix Table A6 reports these estimates. Across alternative definitions of emissions intensity, the average elasticity is $\hat{\varphi}^\star \approx -0.2$. Interpreted heuristically, this estimate implies that investors believe a 1% increase in ESG scores is associated with a 0.2% decline in auto ABS emissions intensity. By contrast, the realized relationship is $\varphi^0 \approx 0.09$. I therefore set $\varphi_B^\star/\varphi^0 \approx -2.25$, corresponding to Point B in Figure 3. Under $\hat{\delta} \approx -0.51$, this implies $\gamma^\star \approx 0.23$.

If investors use ESG scores to price otherwise unobserved collateral-pool CO_2 emissions, the estimates suggest an intended effect of about 0.2% higher issuance spreads for a 1% increase in emissions intensity. Because auto ABS collateral-pool CO_2 emissions are not directly observable, investors may rely on issuer ESG scores as a proxy when allocating capital based on non-pecuniary preferences for greener assets. This interpretation, however, requires subjective beliefs about the ESG-emissions relationship that differ sharply from the realized relationship in the data. Under those beliefs, a 1% higher ESG score corresponds to a 0.2% decrease in emissions intensity. In the data, by contrast, a 1% higher ESG score is associated with a 0.09% increase in emissions intensity.

4.7 Robustness Tests

The result that issuers with high ESG scores that finance high-emission auto loan securitizations have a lower cost of capital is robust to using alternative measures of greenness, tranches, specifications, and estimators. I find similar results using individual ESG scores and other measures of collateral greenness such as the average MPG of vehicles, share of trucks, and an independently

[27] Across sectors, U.S. companies analyzed in a recent S&P Global (2024) study are targeting an average 51% reduction in Scope 1 and 2 emissions as part of their "net zero" plans. I use this figure to operationalize "net zero."

constructed greenness measure by the Kroll Bond Rating Agency (KBRA, 2022). The results continue to hold when excluding deals with a high share of subprime loans. I find quantitatively similar results using propensity score matching and doubly robust machine learning estimators. Appendix Section A.3 provides a detailed discussion of the robustness checks.

5 Pass-through of ESG Investing to Consumer Loan Rates

The impact of ESG investing on consumer loan demand depends on the pass-through of issuance spreads in the ABS market to consumer rates and the elasticity of consumer loan demand with respect to these rates. The percentage change in consumer loan demand is

$$\partial \log \text{Loan Demand} = \underbrace{\frac{\partial \log \text{Loan Demand}}{\partial \log \text{Consumer Rate}}}_{\text{Price Elasticity}} \times \underbrace{\frac{\partial \log \text{Consumer Rate}}{\Delta \text{ ABS Spreads}}}_{\text{Pass-through Semi-Elasticity}} \times \Delta \text{ ABS Spreads}, \quad (10)$$

in which the first term on the right-hand side is the price elasticity of consumer loan demand with respect to consumer interest rates, the second term is the pass-through semi-elasticity of issuance spreads to consumer interest rates, and the last term is the change in ABS issuance spreads.

Changes in ABS Spreads Table 6 reports that auto ABS of issuers with high ESG scores enjoyed 9 bps lower issuance spreads on average from 2017 to 2022. These magnitudes may seem small and their impact on real quantities may appear negligible. However, the effect of changes in ABS spreads on consumer interest rates depends on the pass-through elasticity of changes in ABS spreads to changes in consumer interest rates and on the price elasticity of consumer loan demand with respect to these rates. It is not immediately clear whether an ESG basis spread of 9 bps in the senior tranches of auto ABS will result in a 9 bps change in consumer rates. The ESG basis spread represents a decrease in the average cost of funding for the safest tranche in a diversified pool of thousands of loans, rather than the marginal decrease in funding cost for a single loan. Moreover, manufacturers with captive lenders jointly optimize lending and vehicle sales, which further complicates the pass-through (Benmelech et al., 2017, Benetton et al., 2021, Hankins et al., 2026). Manufacturers often use the vertical integration of credit provision to increase their sales by offering subsidized interest rates to consumers. Below, I show empirically that loan subsidies create an important non-linearity in pass-through.

Estimating the Pass-through to Consumer Rates I estimate the pass-through of the issuance spread of ABS o to consumer loan i in month t using specifications of the form:

$$\log\left(\text{Consumer Loan Interest Rate}\right)_{i,t} = \beta \times \widehat{\text{ABS Spread}}_{o,t-1} + \textit{HDFE} + \mathbf{X}_i'\zeta + \varepsilon_{i,t} \tag{11}$$

The coefficient of interest, β, measures the pass-through elasticity of ABS spreads to consumer loan interest rates in percent. The relationship between ABS spreads and consumer interest rates is obviously endogenous and determined by equilibrium conditions that connect the two markets through a financial intermediary (Benetton et al., 2021). To address this endogeneity, I employ an instrumental variable approach using exogenous shifters of funding costs for auto loan lenders. I use two different instruments as funding cost shifters for auto loan lenders: (i) ICE BofA U.S. Corporate Index Option-Adjusted Spread and (ii) the leave-one-out mean of auto ABS spreads issued in the same month (excluding the originator itself). The idea behind these two instruments is to exploit common variation in funding markets.[28]

The specifications include a set of high-dimensional fixed effects (*HDFE*) that identify the pass-through elasticity using loans with the same characteristics made either *across-issuer but within-time* or *across-time but within-issuer*. To be specific, the *within-time across issuer* specifications include (origination month $\times$ *HDFE Set*) fixed effects. The *within-issuer across time* specifications include (originator $\times$ *HDFE Set*) fixed effects. The *HDFE Set* is given by

$$\textit{HDFE Set}: \left\{ \begin{array}{l} \text{borrower state} \times \text{vehicle type} \times \text{vehicle used} \times \text{loan term quartile} \\ \times \text{LTV quartile} \times \text{warehousing quartile} \times \text{credit score bin} \end{array} \right\}$$

in which credit score bins have a width of 50. The groups defined by *within-time across-issuer* and *within-issuer across-time* FE have on average 24 and 497 observations, respectively. The specifications further include a vector of controls at the loan level, $\mathbf{X}_i$, that linearly control for log-transformation of maturity-matched estimates of the real yield curve, LTV ratio, payment-to-income ratio, loan term, warehousing time, vehicle value, and vehicle age. The standard errors are clustered at collateral pool and origination month.

Table 9 presents the estimates of the pass-through elasticity of auto ABS spreads to consumer interest rates. Several points are noteworthy. First, OLS estimates exhibit an upward bias compared

[28] See Berry and Haile (2021) who write, "Noisy measures of a producer's actual cost shifters can also serve as instruments. For example, the average wage level in a producer's labor market may not perfectly track the producers' labor costs but is nonetheless likely to be highly correlated with those costs. Thus, such wage measures can serve as instruments as long as they are uncorrelated with demand shocks conditional on the exogenous variables [...]"

Table 9: Estimates of the Pass-Through Elasticity from Issuance Spreads to Consumer Loan Rates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	within time across issuers				within issuer across time			
	Dependent variable: Log(Consumer Loan Rate)							
	OLS		IV		OLS		IV	
Spread t-1	0.517*	0.461*	0.194	0.016	0.021	0.004	-0.266*	0.109+
	(0.221)	(0.177)	(0.289)	(0.313)	(0.054)	(0.050)	(0.129)	(0.058)
Spread t-1 × Captive	1.740***	1.670***	0.906*	0.707*	1.219***	1.286***	0.970***	0.654***
	(0.264)	(0.230)	(0.342)	(0.319)	(0.193)	(0.189)	(0.281)	(0.120)
Total effect of Spread t-1 for Captive	2.256***	2.131***	1.100*	0.723	1.240***	1.290***	0.704*	0.763***
	(0.267)	(0.241)	(0.452)	(0.445)	(0.193)	(0.188)	(0.294)	(0.128)
Origination Month × HDFE Set FE	✓	✓	✓	✓				
Originator × HDFE Set FE					✓	✓	✓	✓
Time × State FE					✓	✓	✓	✓
Linear Controls		✓	✓	✓		✓	✓	✓
Instrument			BofA	Others			BofA	Others
Kleibergen-Paap F-stat.			7.50	5.16			10.38	110.00
Adj. R^2	0.912	0.920	0.104	0.145	0.891	0.900	0.090	0.132
Sample	7,585,973	7,546,292	7,546,292	5,700,055	8,852,140	8,804,436	8,804,428	6,644,862

Notes: This table reports regression estimates of the pass-through semi-elasticity from auto ABS markets to consumer interest rates using specifications of the form given in Eq. (11). IV is estimated using Limited Information Maximum Likelihood (LIML) to account for potentially weak instruments. Standard errors in parentheses are double-clustered at the collateral-pool and origination-month levels. * p<0.10, ** p<0.05, *** p<0.01.

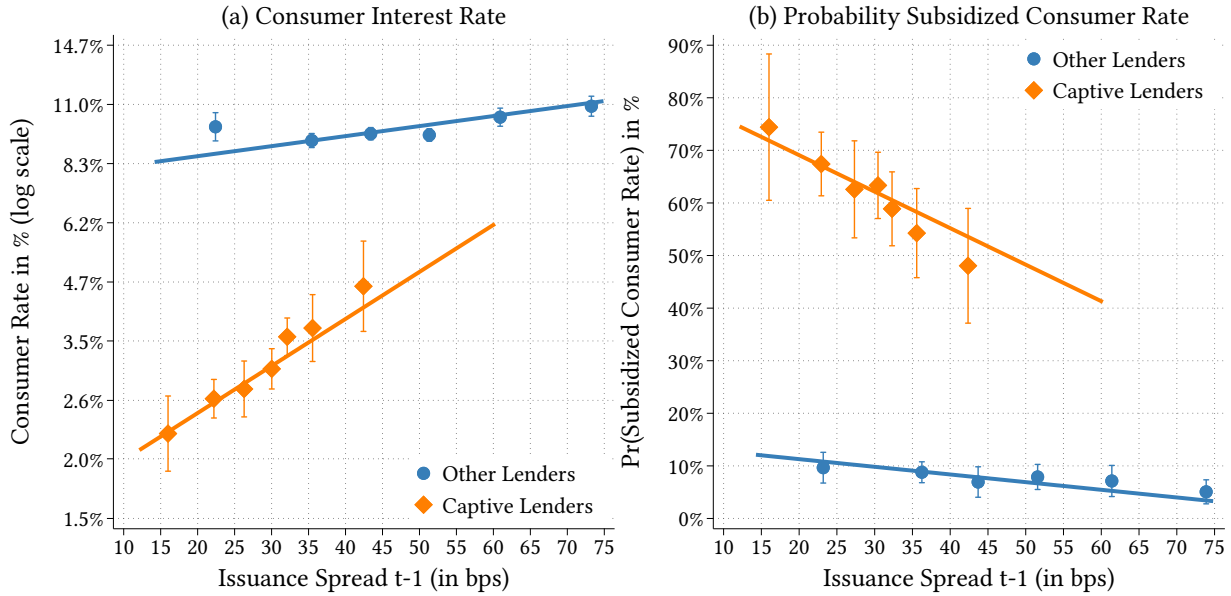
to IV estimates. Second, the estimates for captive lenders are larger than those for non-captive lenders. In IV specifications, the pass-through elasticity is essentially zero for non-captive lenders but large and statistically significant for captive lenders.

Using within-time, across-issuer variation, the estimates in columns (3) and (4) imply pass-through semi-elasticities of 1.10 and 0.72, respectively. Using within-issuer, across-time variation, the estimates in columns (7) and (8) are 0.70 and 0.76. The average loan interest rate by captive lenders is 3.34%. The pass-through estimates thus imply that the ESG basis spread of 9 bps (Table 6) translates into consumer interest rate reductions of 21 bps to 33 bps, on average.

The vertical integration of manufacturing and credit provision drives the large pass-through of the ESG convenience yield by captive lenders. Captive lenders frequently subsidize loans to increase car sales (Benetton et al., 2021). The most common form of subsidy is a reduced interest rate. Captive lenders often advertise 0% or 1.99% financing for new vehicles. Over 66% of captive lenders' subsidized loans in my sample have interest rates less than 2%.

Figure 4 shows that captive lenders increase their supply of subsidized loans when auto ABS issuance spreads are low. A 9 bps decrease in issuance spreads is associated with a 7 percentage point increase in the probability of receiving a subsidized interest rate from a captive lender. The

Figure 4: Relationship between Consumer Interest Rates and Auto ABS Issuance Spreads



Notes: This figure shows binned scatter plots of issuance spreads against consumer rates in Panel (a) and the probability of receiving a subsidized loan in Panel (b) separately for captive and non-captive lenders. The specifications include high-dimensional fixed effects that identify the relationships using loans with similar characteristics made by different lenders during the same month. Variables are trimmed at the 2.5% and 97.5% percentiles. 95% confidence intervals are based on standard errors clustered at the collateral pool and origination month levels.

average difference in interest rates between a subsidized and a non-subsidized loan from a captive lender is 328 bps. The increased probability of receiving a subsidized loan thus lowers consumer interest rates by 28 bps in expectation.

Price Elasticity I rely on the extensive literature on the price elasticity of consumer vehicle loan demand with respect to interest rates instead of directly estimating the elasticity. Argyle et al. (2020) report estimates for the price elasticity of −0.18, with estimates by FICO subgroup ranging from −0.22 to −0.07. Lukas (2017) estimate a loan price elasticity of −0.34. Attanasio, Koujianou Goldberg, and Kyriazidou (2008) report elasticity estimates ranging from −0.09 to −0.82 but cannot reject the null of zero elasticity. Given the considerable range of estimates for intensive-margin price elasticities, I report results for elasticities from −0.18 to −0.5.

Implied Changes in Consumer Loan Demand Table 10 shows the implied changes in consumer loan demand for captive lenders associated with the ESG convenience yield. I provide a range of estimates based on the estimates of price elasticity of consumer loan demand and pass-through elasticity. The implied percentage changes in loan demand range from 1.1% to 4.9%. To illustrate, consider a $33,000 loan for a vehicle with a 3.34% interest rate. The results in column (1) imply that a 9 bps decrease in the auto ABS spread would result in a 1.1% increase in equilibrium

Table 10: Implied Changes in Individual Consumer Loan Demand for Captive Lenders

		(1)	(2)	(3)	(4)	(5)	(6)
Intensive Margin Price Elasticity	$\frac{\partial \log \text{Loan demand}}{\partial \log \text{Consumer rate}}$	−0.18		−0.34		−0.50	
Pass-through Semi-Elasticity	$\frac{\partial \log \text{Consumer rate}}{\Delta \text{ ABS spread in \%}}$	0.70	1.10	0.70	1.10	0.70	1.10
Resulting Change in Loan Demand	$\partial \log$ Loan demand	1.13%	1.78%	2.14%	3.37%	3.15%	4.95%
	Δ Loan demand in USD	\$374	\$588	\$707	\$1,111	\$1,040	\$1,634

Notes: This table reports estimates of the implied change in consumer loan demand for captive lenders with high ESG scores: $\partial \log \text{Loan Demand} = \frac{\partial \log \text{Loan Demand}}{\partial \log \text{Consumer Rate}} \times \frac{\partial \log \text{Consumer Rate}}{\Delta \text{ABS Spreads}} \times \Delta \text{ABS Spreads}$. The average change in ABS spread due to ESG pricing (ESG basis spread) is −9 bps (see Table 6). Intensive-margin price elasticities of consumer vehicle credit demand are from Argyle et al. (2020), Lukas (2017), and Attanasio et al. (2008). The average loan amount for captive lenders is approximately \$33,000.

loan demand, equivalent to approximately \$374. Column (6) implies a change of about \$1,634.

Changes in individual consumer loan demand do not directly equate to changes in vehicle demand. The estimated changes in loan demand in Table 10 are best understood as intensive-margin changes affecting the loan amount for a given vehicle model purchase. This additional loan demand may be used for upgrades or accessories rather than for a higher-priced vehicle model. However, manufacturers and captive lenders benefit from the cumulative increase in loan demand across all loans they extend, leading to a meaningful increase in product demand and profits for manufacturers with high ESG scores.

Who Benefits from Lower ABS Spreads? Lastly, I estimate which buyers benefit from lower auto ABS issuance spreads by estimating the following specification:

$$\begin{aligned} \text{Outcome}_{i,t} = \beta_0 \text{ ABS Spread}_{o,t-1} &+ \beta_1 \text{ Captive}_o \times \text{ABS Spread}_{o,t-1} \\ &+ \beta_2 \text{ Truck}_i \times \text{ABS Spread}_{o,t-1} \\ &+ \beta_3 \text{ Captive}_o \times \text{Truck}_i \times \text{ABS Spread}_{o,t-1} \\ &+ \textit{HDFE} + \mathbf{X}_i'\zeta + \varepsilon_{i,t} \end{aligned} \tag{12}$$

where $\text{Outcome}_{i,t}$ is either the consumer loan interest rate or an indicator variable equal to one if the loan interest rate is subsidized and zero otherwise. Truck is an indicator variable equal to one if the vehicle is a truck and zero otherwise. Captive is an indicator variable equal to one if the lender is a captive auto lender and zero otherwise. Controls and fixed effects are as in Eq. (11).

Table 11 shows that buyers of trucks who finance their purchases with captive lenders benefit

Table 11: Pass-Through to Consumer Loan Pricing by Lender and Vehicle Segment

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Consumer Interest Rate (bps)				Pr(Subsidized Interest Rate) in %			
ABS Spread$_{t-1}$ (bps)	9.357***	0.778**	7.992***	0.813*	-0.047*	0.032	-0.053	0.036
	(1.354)	(0.373)	(0.900)	(0.383)	(0.024)	(0.022)	(0.047)	(0.041)
Captive × ABS Spread$_{t-1}$	-1.538	-0.325	-2.438	-0.234	-0.130	-0.120	0.168	-0.009
	(1.249)	(0.629)	(1.691)	(0.628)	(0.218)	(0.106)	(0.253)	(0.123)
Truck × ABS Spread$_{t-1}$	-1.387***	0.413*	-3.014***	0.222	-0.076*	-0.017	-0.034	-0.019
	(0.443)	(0.246)	(0.479)	(0.203)	(0.041)	(0.025)	(0.046)	(0.031)
Captive × Truck × ABS Spread$_{t-1}$	**6.510*** **	**2.881*** **	5.700	0.143	**-0.889****	**-1.628*** **	0.340	-0.495
	(1.630)	**(0.673)**	(3.285)	(1.240)	**(0.434)**	**(0.291)**	(0.486)	(0.462)
Captive × ESG Score × ABS Spread$_{t-1}$			-0.508	-1.370			**-0.709****	-0.416
			(1.689)	(1.155)			**(0.254)**	(0.273)
Captive × Truck × ESG Score × ABS Spread$_{t-1}$			**7.201****	**6.586*** **			**-2.545*** **	**-2.260*** **
			(2.628)	**(1.772)**			**(0.489)**	**(0.519)**
Other Interaction Terms	✓	✓	✓	✓	✓	✓	✓	✓
Linear Controls	✓	✓	✓	✓	✓	✓	✓	✓
Origination Month × HDFE Set FE	✓		✓		✓		✓	
Originator × HDFE Set FE		✓		✓		✓		✓
Year × State FE		✓		✓		✓		✓
Adj. R^2	0.864	0.871	0.870	0.871	0.614	0.614	0.642	0.640
Observations	7,648,117	7,561,330	6,942,381	6,865,758	8,388,986	8,300,224	7,568,551	7,490,488

Notes: This table reports OLS estimates of Eq. (12). Columns (1) to (4) report estimates of the impact of ABS issuance spreads (in basis points) on consumer interest rates (in basis points). Columns (5) to (8) report estimates of the impact of ABS issuance spreads (in basis points) on receiving a subsidized interest rate in a linear probability model. Truck is an indicator variable equal to one if the vehicle is a truck and zero otherwise. Captive is an indicator variable equal to one if the lender is a captive auto lender of a manufacturer and zero otherwise. Average ESG scores are standardized to have mean zero and standard deviation one. The specifications contain the full set of interaction terms but I report only coefficients of interest for brevity. Odd columns use within-time across-issuer variation, while even columns use within-issuer across-time variation. Standard errors in parentheses are double-clustered at the collateral-pool and origination-month levels. * p<0.10, ** p<0.05, *** p<0.01.

from lower ABS issuance spreads relative to buyers of passenger cars. Columns (1) and (5) use within-time across-issuer variation and show that a 1 bps decrease in issuance spread is associated with a 6.5 bps decrease in consumer interest rates and a 0.9 percentage point increase in the probability of receiving a subsidized interest rate for truck buyers financing with captive lenders relative to buyers of passenger cars. Columns (2) and (6) use within-issuer across-time variation and show that a 1 bps decrease in issuance spread is associated with a 2.9 bps decrease in consumer interest rates and a 1.6 percentage point increase in the probability of receiving a subsidized interest rate for truck buyers financing with captive lenders.

Next, I add interaction terms between issuer-level ESG scores (averaged over data providers) and the key variables in Eq. (12). This allows me to test whether reductions in ABS issuance spreads associated with higher ESG scores are passed through to consumer loan rates and whether this effect differs by vehicle type. I standardize ESG scores to have mean zero and standard deviation one, so the interaction coefficients can be interpreted as the incremental pass-through for an

issuer with a one standard deviation higher ESG score.

The lower interest rates and higher likelihood of subsidized financing for truck buyers relative to buyers of passenger cars borrowing from captive lenders are concentrated among captive lenders with higher ESG scores. Columns (3) and (7), which exploit across-issuer within-time variation, show that an 9 bps ESG basis spread is associated with a 57 bps lower consumer interest rate and a 20 percentage point increase in the probability of receiving a subsidized rate for truck buyers financing with captive lenders whose ESG score is one standard deviation above the mean. Similarly, columns (4) and (8), which exploit within-issuer across-time variation, show that an 9 bps ESG basis spread is associated with a 52 bps lower consumer interest rate and an 18 percentage point increase in the probability of receiving a subsidized rate for truck buyers financing with captive lenders whose ESG score is one standard deviation above the mean.

6 Discussion

I document that ESG investing lowers the cost of capital for auto ABS of issuers with high issuer-level ESG scores. I estimate that investors earn an ESG convenience yield of 0.39% p.a. on their ESG investments. This ESG convenience yield generates seigniorage for issuers of ESG assets and lowers their borrowing cost. Consumers who finance high-emission vehicles with loans from captive lenders benefit from the ESG convenience yield through lower borrowing costs. However, my findings also show that ESG investors do not necessarily invest in the most environmentally friendly securities. Auto ABS investors reward issuers with higher ESG scores with a lower cost of capital, even if their securities have higher CO_2 emissions intensity. The market's focus on issuer-level ESG scores, rather than the collateral's CO_2 emissions, lowers the cost of capital for high-emission vehicles. This raises questions about the effectiveness of ESG investment strategies in addressing environmental externalities.

These findings suggest a need for greater clarity and transparency in ESG labeling and investment processes. ESG fund managers may need to re-evaluate their investment processes to ensure they promote environmentally sustainable investing. Policymakers should provide more guidance to the financial sector on what constitutes environmentally sustainable investing and on how to ensure that ESG labels accurately reflect the environmental impact of investments. The demand for such guidance is consistent with evidence that institutional investors actively seek better climate risk information. Ilhan, Krueger, Sautner, and Starks (2023) find that climate-conscious institutional investors value and demand climate risk disclosures at the firm level. However, my findings suggest that comparable disclosure efforts have not extended to the collateral underlying

securitized products. Investors in the auto ABS market did not have ready access to collateral-pool CO_2 emissions data during my sample period, which likely contributed to their reliance on issuer-level ESG scores and the resulting misallocation of capital toward high-emission securitizations.

ESG regulation in the U.S. is still in its infancy. The SEC has moved toward requiring that ESG labels accurately reflect the environmental impact of investments and that funds disclose their ESG practices transparently, though these rules have not been finalized. In Europe, similar efforts are underway with the adoption of the Sustainable Finance Disclosure Regulation (SFDR), which requires comprehensive and transparent disclosure of sustainability objectives. Emiris, Harris, and Koulischer (2023) examine the impact of the SFDR on ESG flows. They find that the regulation increased flows to ESG funds, particularly from environmentally conscious investors, and that funds with higher initial uncertainty about their sustainability benefited most from the disclosure.

7 Conclusion

Many ESG investors want to raise the cost of emitting CO_2 by rewarding "green" assets with a lower cost of capital and penalizing "brown" assets with a higher cost of capital. This paper shows that ESG investing lowers the cost of capital for auto ABS issuers with high ESG scores. The pass-through of this green convenience yield to consumer interest rates can be significant for captive lenders, resulting in economically meaningful changes in loan demand.

However, the market's focus on issuer-level ESG scores, rather than the collateral's CO_2 emissions, also lowers the cost of capital for high-emission securitizations because ESG scores correlate positively with emissions. Consequently, investors who rely on ESG scores to allocate capital inadvertently subsidize CO_2 emissions. ESG mutual funds allocate more capital to auto ABS from issuers with high ESG scores even if those deals finance high-emission vehicles. A model of subjective beliefs in which ESG investors use a representativeness heuristic to infer CO_2 emissions from firm-level ESG scores can explain the observed effects.

These findings highlight that while ESG investing can have a meaningful impact, it does not necessarily increase the cost of emitting CO_2. The findings also underscore the need for accurate and comprehensive project-level ESG metrics that reflect environmental impact.

References

Abadie, A. and G. W. Imbens (2016): "Matching on the Estimated Propensity Score," *Econometrica*, 84, 781–807. [Cited on page 48.]

Amel-Zadeh, A. and G. Serafeim (2018): "Why and How Investors Use ESG Information: Evidence from a Global Survey," *Financial Analysts Journal*, 74, 87–103. [Cited on page 3.]

Ardia, D., K. Bluteau, K. Boudt, and K. Inghelbrecht (2023): "Climate change concerns and the performance of green vs. brown stocks," *Management Science*, 69, 7607–7632. [Cited on pages 5, 20, 21, 22, 23, 24, 43, and 45.]

Argyle, B. S., T. D. Nadauld, and C. J. Palmer (2020): "Monthly Payment Targeting and the Demand for Maturity," *The Review of Financial Studies*, 33, 5416–5462. [Cited on pages 7, 31, and 32.]

Aswani, J. and S. Rajgopal (2025): "Rethinking the value and emission implications of green bonds," *Management Science*, forthcoming. [Cited on page 8.]

Attanasio, O. P., P. Koujianou Goldberg, and E. Kyriazidou (2008): "Credit Constraints in the Market for Consumer Durables: Evidence from Micro Data on Car Loans," *International Economic Review*, 49, 401–436. [Cited on pages 31 and 32.]

Avramov, D., A. Lioui, Y. Liu, and A. Tarelli (2024): "Dynamic ESG Equilibrium," *Management Science*. [Cited on pages 5 and 20.]

Baker, M., D. Bergstresser, G. Serafeim, and J. Wurgler (2022): "The Pricing and Ownership of US Green Bonds," *Annual Review of Financial Economics*, 14, 415–437. [Cited on page 8.]

Belloni, A., V. Chernozhukov, and C. Hansen (2014): "Inference on Treatment Effects After Selection Among High-Dimensional Controls," *Review of Economic Studies*, 81, 608–650. [Cited on pages 17, 48, and 55.]

Bena, J., B. Bian, and H. Tang (2023): "Financing the Global Shift to Electric Mobility," Working paper, Available at SSRN 4526150. [Cited on page 10.]

Benetton, M., S. Mayordomo, and D. Paravisini (2021): "Credit Fire Sales: Captive Lending As Liquidity in Distress," Working paper, Available at SSRN 3780413. [Cited on pages 7, 9, 28, 29, and 30.]

Benmelech, E., R. R. Meisenzahl, and R. Ramcharan (2017): "The Real Effects of Liquidity During the Financial Crisis: Evidence from Automobiles," *The Quarterly Journal of Economics*, 132, 317–365. [Cited on pages 9, 12, and 28.]

Berg, F., K. Fabisik, and Z. Sautner (2020): "Is History Repeating Itself? The (Un)Predictable Past of ESG Ratings," Finance Working Paper 708/2020, European Corporate Governance Institute. [Cited on page 8.]

Berg, F., J. F. Koelbel, A. Pavlova, and R. Rigobon (2022a): "ESG Confusion and Stock Returns: Tackling the Problem of Noise," Working paper 30562, National Bureau of Economic Research. [Cited on page 9.]

Berg, F., J. F. Koelbel, and R. Rigobon (2022b): "Aggregate Confusion: The Divergence of ESG Ratings," *Review of Finance*, 26, 1315–1344. [Cited on pages 8 and 13.]

Berk, J. and J. H. van Binsbergen (2025): "The Impact of Impact Investing," *Journal of Financial Economics*, 164. [Cited on page 8.]

Berry, S. T. and P. A. Haile (2021): "Foundations of Demand Estimation," in *Handbook of Industrial Organization*, Elsevier, vol. 4, 1–62. [Cited on page 29.]

Broccardo, E., O. Hart, and L. Zingales (2022): "Exit Versus Voice," *Journal of Political Economy*, 130, 3101–3145. [Cited on page 6.]

Buberger, J., A. Kersten, M. Kuder, R. Eckerle, T. Weyh, and T. Thiringer (2022): "Total CO2-Equivalent Life-Cycle Emissions from Commercially Available Passenger Cars," *Renewable and Sustainable Energy Reviews*, 159, 112158. [Cited on page 3.]

Campbell, J. Y. and R. J. Shiller (1988): "The Dividend-Price Ratio and Expectations of Future Dividends and Discount Factors," *The Review of Financial Studies*, 1, 195–228. [Cited on page 15.]

Chava, S. (2014): "Environmental Externalities and Cost of Capital," *Management Science*, 60, 2223–2247. [Cited on page 8.]

DeMarzo, P. M. (2005): "The Pooling and Tranching of Securities: A Model of Informed Intermediation," *The Review of Financial Studies*, 18, 1–35. [Cited on page 2.]

Edmans, A., D. Levit, and J. Schneemeier (2022): "Socially Responsible Divestment," Working Paper 823, European Corporate Governance Institute. [Cited on page 6.]

Emiris, M., J. Harris, and F. Koulischer (2023): "The Effect of Environmental Preferences on Investor Responses To ESG Disclosure," Working paper, University of Chicago. [Cited on page 35.]

Eskildsen, M., M. Ibert, T. I. Jensen, and L. H. Pedersen (2024): "In Search of the True Greenium," Working paper, Available At SSRN 4744608. [Cited on page 5.]

Filipović, D., M. Pelger, and Y. Ye (2025): "Stripping the Discount Curve—a Robust Machine Learning Approach," *Management Science*. [Cited on pages 9, 16, 19, and 56.]

Flammer, C. (2021): "Corporate Green Bonds," *Journal of Financial Economics*, 142, 499–516. [Cited on page 8.]

Gennaioli, N. and A. Shleifer (2010): "What Comes to Mind," *The Quarterly Journal of Economics*, 125, 1399–1433. [Cited on pages 6, 26, and 27.]

Gillan, S. L., A. Koch, and L. T. Starks (2021): "Firms and Social Responsibility: A Review of ESG and CSR Research in Corporate Finane," *Journal of Corporate Finance*, 66, 101889. [Cited on page 8.]

Gorton, G. (2017): "The History and Economics of Safe Assets," *Annual Review of Economics*, 9, 547–586. [Cited on page 4.]

Goss, A. and G. S. Roberts (2011): "The Impact of Corporate Social Responsibility on the Cost of Bank Loans," *Journal of Banking & Finance*, 35, 1794–1810. [Cited on page 8.]

Haber, S. H., J. D. Kepler, D. F. Larcker, A. Seru, and B. Tayan (2022): "2022 Survey of Investors, Retirement Savings, and ESG," Technical report, Stanford Graduate School of Business. [Cited on pages 6, 26, and 27.]

Hankins, K. W., M. Momeni, and D. Sovich (2026): "Consumer Credit and the Incidence of Tariffs: Evidence from the Auto Industry," *American Economic Review*, 116, 627–673. [Cited on pages 9 and 28.]

Hartzmark, S. M. and K. Shue (2026): "Counterproductive Impact Investing: The Impact Elasticity of Brown and

Green Firms," *The Journal of Finance*, forthcoming. [Cited on page 8.]

Hartzmark, S. M. and A. B. Sussman (2019): "Do Investors Value Sustainability? A Natural Experiment Examining Ranking and Fun Flows," *The Journal of Finance*, 74, 2789–2837. [Cited on pages 21 and 50.]

He, A. and B. Mizrach (2017): "Analysis of Securitized Asset Liquidity," *Research Note, FINRA Office of the Chief Economist*. [Cited on page 12.]

Heinkel, R., A. Kraus, and J. Zechner (2001): "The Effect of Green Investment on Corporate Behavior," *Journal of Financial and Quantitative Analysis*, 36, 431–449. [Cited on page 8.]

Hirschman, A. O. (1970): *Exit, Voice and Loyalty: Responses to Decline in Firms, Organizations, and States*, Harvard University Press, Cambridge. [Cited on page 6.]

Hong, H. and E. P. Shore (2023): "Corporate Social Responsibility," *Annual Review of Financial Economics*. [Cited on page 8.]

Ilhan, E., P. Krueger, Z. Sautner, and L. T. Starks (2023): "Climate risk disclosure and institutional investors," *The Review of Financial Studies*, 36, 2617–2650. [Cited on page 34.]

Kacperczyk, M. and J.-L. Peydró (2021): "Carbon Emissions and the Bank-Lending Channel," Finance working paper no. 991/2024, European Corporate Governance Institute. [Cited on page 5.]

KBRA (2022): "ESG: Auto Loan ABS: EPA Greenhouse Gas Score: Second-Quarter 2022," . [Cited on pages 28 and 47.]

Klee, E., A. Morse, and C. Shin (2023): "Auto Finance in the Electric Vehicle Transition," Working paper, Available at SSRN 4642783. [Cited on page 10.]

Klee, E. and C. Shin (2020): "Post-Crisis Signals in Securitization: Evidence from Auto ABS," Working Paper FEDS 2020-042, Board of Governors of the Federal Reserve System. [Cited on pages 9 and 16.]

Koijen, R. S. and M. Yogo (2019): "A Demand System Approach To Asset Pricing," *Journal of Political Economy*, 127, 1475–1515. [Cited on page 50.]

Krishnamurthy, A. and A. Vissing-Jorgensen (2012): "The Aggregate Demand for Treasury Debt," *Journal of Political Economy*, 120, 233–267. [Cited on pages 4, 8, and 14.]

Lukas, M. (2017): "Estimating Interest Rate Elasticities in Consumer Credit," *Economics Letters*, 156, 155–158. [Cited on pages 31 and 32.]

Oehmke, M. and M. M. Opp (2024): "A Theory of Socially Responsible Investment," *Review of Economic Studies*, rdae048. [Cited on page 8.]

Pástor, L., R. F. Stambaugh, and L. A. Taylor (2021): "Sustainable Investing in Equilibrium," *Journal of Financial Economics*, 142, 550–571. [Cited on page 8.]

——— (2022): "Dissecting Green Returns," *Journal of Financial Economics*, 146, 403–424. [Cited on page 8.]

Pastor, L., R. F. Stambaugh, and L. A. Taylor (2025): "Green tilts," *Journal of Financial Economics*, 174, 104173. [Cited on pages 5 and 50.]

Pedersen, L. H., S. Fitzgibbons, and L. Pomorski (2021): "Responsible Investing: The ESG-Efficient Frontier," *Journal of Financial Economics*, 142, 572–597. [Cited on pages 4 and 16.]

Ramcharan, R., S. Verani, and S. J. Van den Heuvel (2016): "From Wall Street to Main Street: The Impact of the Financial Crisis on Consumer Credit Supply," *The Journal of Finance*, 71, 1323–1356. [Cited on pages 9 and 12.]

Sastry, P. R., E. Verner, and D. Marques-Ibanez (2024): "Business as usual: Bank net zero commitments, lending, and engagement," Working paper, National Bureau of Economic Research. [Cited on page 5.]

Seltzer, L. H., L. Starks, and Q. Zhu (2026): "Climate Regulatory Risk and Corporate Bonds," *Journal of Financial Economics*, forthcoming. [Cited on page 8.]

Siani, K. Y. (2025): "Raising Bond Capital in Segmented Markets," *The Review of Financial Studies*, hhaf092. [Cited on page 23.]

van der Beck, P. (2026): "Flow-Driven ESG Returns," *Journal of Finance*, forthcoming. [Cited on pages 5, 21, 22, 24, 45, and 50.]

Zerbib, O. D. (2019): "The Effect of Pro-Environmental Preferences on Bond Prices: Evidence from Green Bond," *Journal of Banking & Finance*, 98, 39–60. [Cited on page 8.]

A Appendix

A.1 Appendix Figures

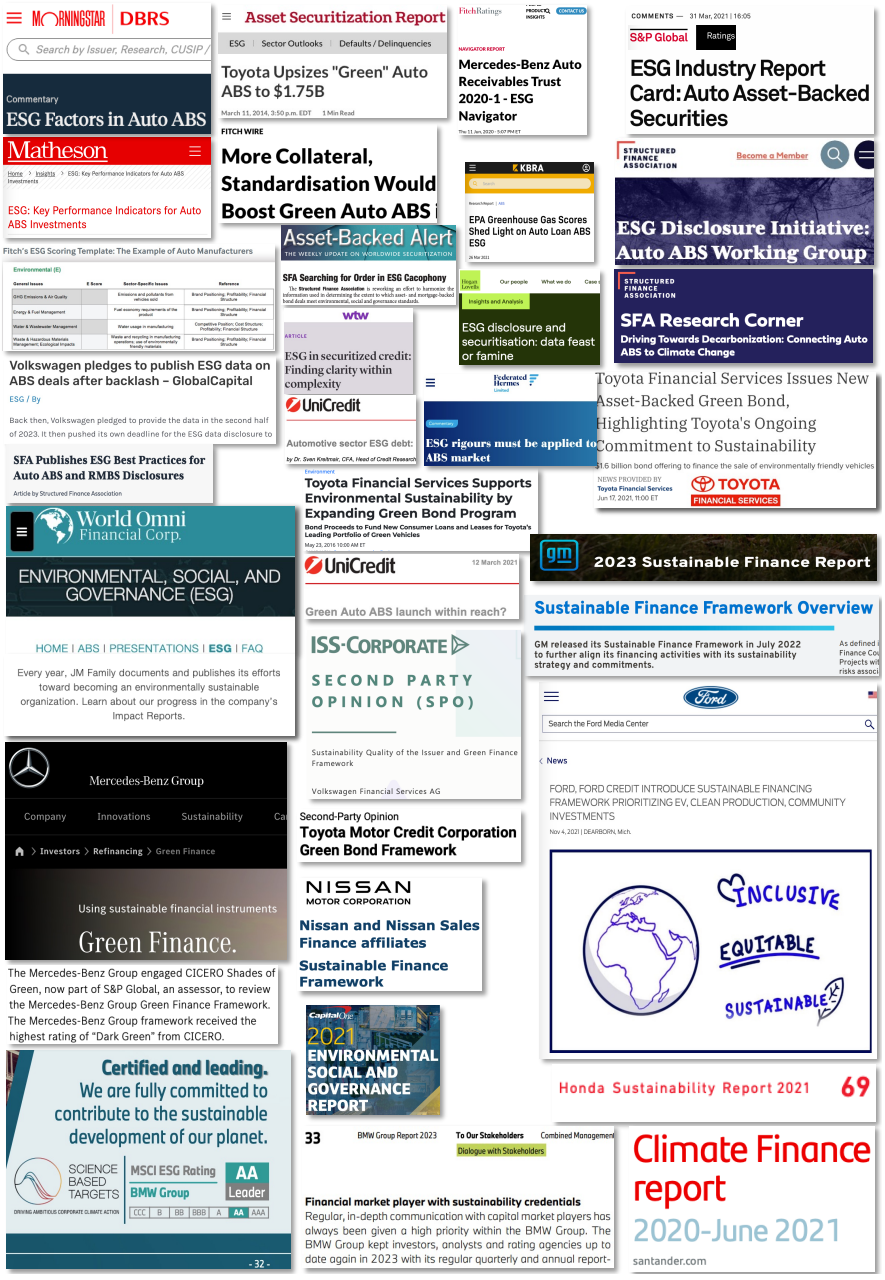


Figure A1: Clippings from ABS Issuers, Investors, Bond Rating Agencies, Industry Associations, Law Firms, and News Outlets that Discuss the Importance of ESG Issues in the Auto ABS Market

A.2 Appendix Tables

Table A1: Regression of average tCO_2 emissions per vehicle on vehicle types

	Constant	Truck share	SUV share	Adj. R^2	N	Avg. tCO_2/vehicle
β	44.108***	1.019***	0.210***	0.746	281	70.514
(se) or sd	(1.879)	(0.036)	(0.040)			15.550

Notes: This table reports coefficients from a regression of average tCO_2 emissions per vehicle on vehicle types. * p<0.10, ** p<0.05, *** p<0.01.

Table A2: Summary Statistics of Loan-Level Data

	Mean	SD	Median	Min	Max	Obs.
Original Interest Rate	7.84	7.00	5.25	0.00	30.00	17,823,551
Original Loan Amount ($)	25,822.58	12,251.91	23,650.84	518.03	248,681.95	17,823,552
Original Loan Term (months)	67.65	8.59	72.00	7.00	96.00	17,823,552
Credit Score	708.64	101.70	719.00	250.00	900.00	17,143,023
Payment-to-Income Share	0.08	0.05	0.08	0.00	0.79	17,700,290
Income Verified	0.09	0.29	0.00	0.00	1.00	17,823,552
Loan-to-Value	0.90	0.16	1.00	0.01	1.00	17,822,211
Outstanding Balance Share	0.83	0.24	0.93	0.00	1.00	17,823,548
Vehicle Value Amount ($)	27,341.46	13,177.32	24,998.00	0.00	1,084,455.00	17,823,549
Vehicle Age (Years)	2.74	2.56	2.00	0.00	35.00	17,823,552
Used Vehicle	0.48	0.50	0.00	0.00	1.00	17,823,552
SVM, Financed	161,660.73	40,008.49	171,346.10	254.15	240,728.61	17,823,552
SVM, Total	202,834.40	16,986.18	207,738.97	189,173.82	240,728.61	17,823,552
tCO2, total Lifetime	78.28	30.61	72.45	0.00	538.75	17,823,552
tCO2, remaining Lifetime	62.12	29.51	56.48	0.00	538.75	17,823,552
tCO2, financed remaining Lifetime	46.57	27.79	44.58	0.00	538.75	17,822,207

Notes: This table reports summary statistics for the loan-level data. Credit scores outside the FICO Auto Score range of 250 to 900 are set to missing.

Table A3: Pricing Results Using Individual ESG Scores

	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread	(4) Issuance Spread	(5) Issuance Spread	(6) Issuance Spread	(7) Issuance Spread	(8) Issuance Spread
Refinitiv ESG Score	-0.525***							
	(0.084)							
Refinitiv Env. Score		-0.0900**						
		(0.040)						
S&P ESG Score			-0.222***					
			(0.048)					
S&P Env. Score				-0.126***				
				(0.032)				
MSCI ESG Score					-0.225***			
					(0.059)			
MSCI Env. Score						-0.198***		
						(0.052)		
Sustainalytics ESG Score							-0.384***	
							(0.123)	
Sustainalytics Env. Score								-0.201***
								(0.054)
Year-month FE, daily market controls	✓	✓	✓	✓	✓	✓	✓	✓
Prepayment speed FE, tranche controls	✓	✓	✓	✓	✓	✓	✓	✓
Adj. R^2	0.98	0.96	0.97	0.97	0.98	0.98	0.95	0.95
Observations	243	243	243	243	231	231	122	122

Table A4: The Effects of Media Climate Change Concerns on Spreads of Auto ABS

	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread	(4) Issuance Spread	(5) Issuance Spread	(6) Issuance Spread	(7) Issuance Spread	(8) Issuance Spread
ESG Score	-0.163	-0.215	-0.312**	-0.285**				
	(0.157)	(0.146)	(0.120)	(0.115)				
Aggregate News Index × ESG Score	-0.366*							
	(0.199)							
Societal Debate News Index × ESG Score		-0.266						
		(0.165)						
Business Impact/Transition Risk News Index × ESG Score			-0.149					
			(0.153)					
Car Industry Transition Risk News Index × ESG Score				-0.272				
				(0.157)				
Environmental Score					-0.322	-0.345	-0.441**	-0.480***
					(0.224)	(0.230)	(0.188)	(0.161)
Aggregate News Index × Environmental Score					-0.443*			
					(0.250)			
Societal Debate News Index × Environmental Score						-0.313		
						(0.230)		
Business Impact/Transition Risk News Index × Environmental Score							-0.190	
							(0.221)	
Car Industry Transition Risk News Index × Environmental Score								-0.359
								(0.216)
Year-month FE, daily market controls	✓	✓	✓	✓	✓	✓	✓	✓
Prepayment speed FE, tranche controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-ante prepayment controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-post prepayment controls	✓	✓	✓	✓	✓	✓	✓	✓
Adj. R^2	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98
Observations	243	243	243	243	243	243	243	243

Notes: This table reports elasticity estimates with respect to ESG scores interacted with the Media Climate Change Concerns Index of Ardia et al. (2023). Standard errors in parentheses are clustered at year-month. * p<0.10, ** p<0.05, *** p<0.01.

Table A5: Estimates of δ Using ESG Scores as Instruments For CO_2 Emissions

	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread	(4) Issuance Spread	(5) Issuance Spread	(6) Issuance Spread	(7) Issuance Spread	(8) Issuance Spread
Financed tCO2 per $100,000	-0.441**	-0.470**			-0.512**	-0.567**		
	(0.221)	(0.235)			(0.254)	(0.275)		
Financed tCO2 per Vehicle			-0.468**	-0.488**			-0.550**	-0.609**
			(0.219)	(0.228)			(0.270)	(0.294)
Instruments	ESG & Env.	ESG & Env.	ESG & Env.	ESG & Env.	Env. Scores	Env. Scores	Env. Scores	Env. Scores
Kleibergen-Paap First-stage F-stat.	8.9	8.0	6.3	5.6	16.5	13.1	11.4	9.2
Year-month FE, daily market controls	✓	✓	✓	✓	✓	✓	✓	✓
Prepayment speed FE, tranche controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-ante prepayment controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-post prepayment controls		✓		✓		✓		✓
Observations	231	231	231	231	231	231	231	231

Notes: This table reports estimates of Eq. (6) in which CO_2 emissions are instrumented with individual ESG and environmental pillar scores. Equations are estimated using Limited Information Maximum Likelihood (LIML) to account for potentially weak instruments. All variables are in logs. Observations are weighted by deal size. Standard errors in parentheses are clustered at year-month and originator. * p<0.10, ** p<0.05, *** p<0.01.

Table A6: Elasticity of CO_2 Emissions with respect to ESG Scores at Firm-Level in Compustat

	(1) $\frac{\text{Scope 1 + 2}}{\text{Sales}}$	(2) $\frac{\text{Scope 1 + 2}}{\text{COGS}}$	(3) $\frac{\text{Scope 1 + 2}}{\text{PPEGT}}$	(4) $\frac{\text{Scope 1 + 2}}{\text{Sales}}$	(5) $\frac{\text{Scope 1 + 2}}{\text{COGS}}$	(6) $\frac{\text{Scope 1 + 2}}{\text{PPEGT}}$
Average ESG Score ($\widehat{\varphi}^{\star}$)	-0.304**	-0.293**	-0.313**			
	(0.131)	(0.133)	(0.123)			
Average Env. Score ($\widehat{\varphi}^{\star}$)				-0.110***	-0.063**	-0.087***
				(0.036)	(0.029)	(0.023)
Constant	5.323***	5.897***	5.974***	4.392***	4.882***	4.964***
	(0.512)	(0.521)	(0.481)	(0.095)	(0.072)	(0.063)
NAICS-4 Industry × Year FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.758	0.735	0.492	0.747	0.718	0.487
Observations	5,172	5,121	5,121	5,506	5,503	5,235

Notes: This table shows coefficient estimates from Scope 1+2 CO_2 Emissions Intensity $= \beta_0 + \beta_1$ ESG Score $+ \varepsilon$. Scope 1+2 emissions are from S&P Trucost from 2009 to 2022, excluding estimated data. Sales, cost of goods sold (COGS), and property, plant, and equipment (PPEGT) are from S&P Compustat North America. Sales, COGS, and PPEGT are trimmed at 2.5/97.5% (yearly) to remove outliers. All variables are in logs. Standard errors in parentheses are clustered at year and firm. * p<0.10, ** p<0.05, *** p<0.01.

Table A7: Underwriting Discounts for SBTi-Committed Underwriters

	(1)	(2)	(3)	(4)	(5)
	Underwriting Discount (in logs)				
SBTi Underwriter=1	0.086***	0.072**	0.069**	0.049***	0.049***
	(0.031)	(0.030)	(0.031)	(0.017)	(0.017)
SBTi Underwriter=1 × ESG Capital Flow ($100bn)			0.010*		
			(0.006)		
SBTi Underwriter=1 × Aggregate News Index				0.136**	
				(0.056)	
SBTi Underwriter=1 × Env. Concerns News Index					0.133**
					(0.058)
Year-month, Subprime FE	✓	✓	✓	✓	✓
Other controls		✓	✓	✓	✓
Pseudo R^2	0.04	0.04	0.04	0.10	0.10
Observations	276	276	225	276	276

Notes: This table reports PPML estimates of the log underwriting discount on the SBTi Underwriter indicator and its interactions with ESG fund flows from van der Beck (2026) and the aggregate and environmental concerns news indices of Ardia et al. (2023). The dependent variable is the underwriting discount retained by lead underwriters; column (1) uses the raw discount and columns (2) through (5) use deal-size–weighted discounts. Other controls include log deal size, the number of lead underwriters, tranche controls, and an indicator for whether the issuer is a U.S. manufacturer. All specifications include year-month and subprime fixed effects. Standard errors in parentheses are clustered at originator and year-month. * p<0.10, ** p<0.05, *** p<0.01.

A.3 Robustness Tests

The result that issuers with high ESG scores that finance high-emission auto ABS have a lower cost of capital is robust to using alternative measures, tranches, specifications, and estimators.

Table A8: Elasticity of Issuance Spreads with Respect to Emissions in Prime Auto ABS only

	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread	(4) Issuance Spread	(5) Issuance Spread	(6) Issuance Spread	(7) Issuance Spread	(8) Issuance Spread
Financed tCO2 per $100,000	-0.183*	-0.222*						
	(0.099)	(0.109)						
Expected tCO2 per $100,000			-0.223**	-0.265**				
			(0.083)	(0.097)				
Financed tCO2 per Vehicle					-0.172*	-0.206**		
					(0.086)	(0.096)		
Expected tCO2 per Vehicle							-0.172*	-0.209**
							(0.085)	(0.095)
Year-month FE, daily market controls	✓	✓	✓	✓	✓	✓	✓	✓
Prepayment speed FE, tranche controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-ante prepayment controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-post prepayment controls		✓		✓		✓		✓
Adj. R^2	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97
Observations	190	190	190	190	190	190	190	190

Notes: This table reports estimates of the risk-adjusted pricing model of Eq. (6) using prime auto ABS deals only. Standard errors in parentheses are clustered at year-month and originator. * p<0.10, ** p<0.05, *** p<0.01.

Lower Cost of Capital for Brown Auto ABS is Unrelated to Credit Quality A potential concern is that I use both prime and subprime auto ABS. Differences in CO_2 emissions may be correlated with unobserved characteristics related to loan quality. For example, subprime borrowers more often buy used vehicles and likely find it harder to refinance their loans than prime borrowers. Appendix Table A8 shows that the results still hold when excluding subprime auto ABS. The estimated elasticities of issuance spreads with respect to emissions are between 0.17 and 0.27 in prime auto ABS, similar to the main result in Table 5. This alleviates potential concerns that the unobserved heterogeneity in credit quality contaminates the original estimates.

Table A9: Elasticity of Issuance Spreads with Respect to Different Measures of Greenness

	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread	(4) Issuance Spread	(5) Issuance Spread	(6) Issuance Spread	(7) Issuance Spread	(8) Issuance Spread	(9) Issuance Spread	(10) Issuance Spread
Expected tCO2 per $100,000	-0.199**	-0.222**								
	(0.093)	(0.090)								
Expected tCO2 per Vehicle			-0.164*	-0.175**						
			(0.085)	(0.082)						
Avg. MPG × (-1)					-0.239	-0.182				
					(0.197)	(0.206)				
Avg. Share of Trucks							-0.114	-0.156		
							(0.103)	(0.101)		
Avg. GHG Rating × (-1)									-0.315	-0.226
									(0.207)	(0.215)
Year-month FE, daily market controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Prepayment speed FE, tranche controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Ex-ante prepayment controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Ex-post prepayment controls		✓		✓		✓		✓		✓
Adj. R^2	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.97	0.97
Observations	281	281	281	281	281	281	281	281	247	247

Notes: This table reports estimates of the risk-adjusted pricing model of Eq. (6) with different measures of greenness. Average MPG and GHG Rating are multiplied by (-1) such that higher values are environmentally worse. Standard errors in parentheses are clustered at year-month and originator. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Lower Cost of Capital for Brown Auto ABS is Robust to Alternative Measures I perform a series of robustness tests using different measures of greenness for each auto ABS deal: (i) expected tCO_2 per USD, (ii) expected tCO_2 per vehicle, (iii) average MPG of the vehicles in the collateral pool, (iv) average truck share in the collateral pool, and (v) an independently constructed greenness measure by the Kroll Bond Rating Agency (KBRA).[29]

Appendix Table A9 shows that the results remain qualitatively unchanged when different measures of relative greenness are used. All specifications indicate that browner auto ABS have a lower cost of capital. Quantitatively, most estimates imply an elasticity of approximately -0.2, which is close to the estimates in the main results of Table 5.

Lower Cost of Capital of Brown Auto ABS Holds Across the Capital Structure The main analysis uses A-2 tranches due to their similar characteristics across different deals: low credit risk, non-binding clean-up call options, and the highest observation count. However, the results are robust to the choice of other AAA-rated tranches.

Appendix Table A10 reports results for all AAA-rated tranches, showing qualitatively and quantitatively similar outcomes to the main results in Table 5. The estimated elasticities of issuance spreads with respect to emissions are also close to -0.2 in other tranches, showing that the lower cost of capital scales through the entire capital structure of these deals.

[29] KBRA (2022) maps the EPA's vehicle GHG scores (1 to 10, with higher values indicating lower emissions) to 247 auto ABS. GHG scores have been displayed on window labels of new vehicles in the U.S. since 2013.

Table A10: Elasticity of Issuance Spreads with Respect to Emissions in Other Senior Tranches

	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread	(4) Issuance Spread	(5) Issuance Spread	(6) Issuance Spread	(7) Issuance Spread	(8) Issuance Spread
Financed tCO2 per USD	-0.207**	-0.253***			-0.271***	-0.298***		
	(0.081)	(0.078)			(0.062)	(0.066)		
Financed tCO2 per Vehicle			-0.175**	-0.222***			-0.245***	-0.265***
			(0.077)	(0.073)			(0.048)	(0.056)
Year-month FE, daily market controls	✓	✓	✓	✓	✓	✓	✓	✓
Prepayment speed FE, tranche controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-ante prepayment controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-post prepayment controls		✓		✓		✓		✓
Adj. R^2	0.96	0.96	0.96	0.96	0.98	0.98	0.98	0.98
Observations	230	230	230	230	190	190	190	190

Notes: This table reports estimates of the risk-adjusted pricing model of Eq. (6) in other senior tranches. Standard errors in parentheses are clustered at year-month and originator. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Lower Cost of Capital for Brown and High-ESG Auto ABS is Robust to Different Estimators A potential concern with the main analysis is that OLS estimators may not accurately control for differences in covariates and falsely attribute differences in issuance spreads to differences in greenness. I address this concern using two alternative estimators.

First, I use the Propensity-Score Matching estimator described by Abadie and Imbens (2016). Online Appendix Table B3 shows that the matching estimator results for ESG scores are similar to those from dummy OLS specifications in Appendix Table B2, whereas the matching estimator results for the low-emission indicator are larger. This likely occurs because the matching estimator selects a sample more similar in terms of covariates than the OLS estimator, suggesting that the main results underestimate the effect of CO_2 emissions on issuance spreads.

Second, I use the Double-Lasso estimator from Belloni et al. (2014). Online Appendix Table B4 shows that the Double-Lasso estimator results are similar to the main results in Table 5, even when including over 850 potential control variables. The estimator automatically selects relevant control variables for both the outcome and treatment via Lasso estimation. This procedure involves three steps: (1) selecting controls that predict treatment via Lasso, (2) selecting controls that predict the outcome via Lasso, and (3) estimating treatment effects using linear regression while controlling for the union of the selected variables. This method provides inference that is uniformly valid over a large class of models.

Table A11: Summary Statistics of Mutual Fund Portfolio Data

	Mean	SD	Median	Min	Max	N
Portfolio Share	0.18	0.28	0.09	0.00	4.94	11,474
Coupon Yield	1.95	1.19	1.95	0.00	6.51	11,474
Tranche Size ($m)	263.06	168.81	230.00	8.51	746.94	11,474
Weighted Average Life	2.36	0.99	2.39	0.11	5.06	11,474
Subprime ABS	0.41	0.49	0.00	0.00	1.00	11,474
Financed tCO2 per $100,000	227.20	37.78	226.11	118.20	311.78	11,474
Financed tCO2 per Vehicle	59.39	11.52	58.31	40.54	101.25	11,474

Notes: This table reports summary statistics for the mutual fund portfolio data.

A.4 The Auto ABS Holdings of ESG Mutual Funds

ESG Mutual Funds' Approach to Auto ABS Prospectuses of ESG mutual funds often detail their investment approach to asset-backed securities. For example,

"[...] When evaluating securitized debt securities [...], the Adviser generally considers the issuer's ESG rating along with ESG factors related to the underlying pool of assets, such as energy efficiency and environmental impact of the underlying assets"

– ESG Mutual Fund Prospectus I

or

"[...] Potential asset-backed securities are evaluated according to the manager's assessment of material ESG issues for the ABS sectors. The assessment utilizes sector specific metrics across ESG categories, insights from third-party data providers, our analysts' qualitative assessment [...] Environmental assessment involves issues such as carbon emissions, pollution, and renewable energy"

– ESG Mutual Fund Prospectus II

Mutual Fund Portfolio Data I obtain mutual fund holdings from the SEC Form N-PORT, starting from 2019-Q3 when they first became available. I keep the first observation in which a mutual fund reports a position in a senior tranche of an auto ABS. I identify ESG mutual funds by name using keywords such as "sustainable", "ESG", or "climate" and using a list of "Sustainable Investment Mutual Funds and ETFs" offered by institutional member firms of "The Forum for Sustainable and Responsible Investing."[30] Appendix Table A11 shows summary statistics of the mutual fund holding data.

I identify 35 self-declared ESG funds (and 787 non-ESG funds) that hold at least one position in an

[30]The keyword list contains: "green", "climate", "esg", "sustainable", "environment", "responsible", "impact", "catholic", "social", "sri", "csr", "community", and "justice".

auto ABS tranche over the sample period. Note that this is a lower bound on the actual number of portfolios with an ESG tilt. van der Beck (2026) reports that net flows into portfolios with ESG tilts reached $1.3 trillion in 2022, far exceeding the $350 billion in net flows into labeled ESG mutual funds. Similarly, Pastor et al. (2025) estimate that the typical institution's ESG tilt has grown from 17% to 27% between 2012 and 2023. Even mutual funds without declared ESG objectives are affected by marketwide ESG concerns: Hartzmark and Sussman (2019) use the introduction of Morningstar's sustainability ratings to show that "[b]eing categorized as low sustainability resulted in net outflows of more than $12 billion while being categorized as high sustainability led to net flows of more than $24 billion."

Identification Strategy I estimate a reduced form asset demand system in the spirit of Koijen and Yogo (2019) to test whether ESG funds tilt their portfolios toward greener auto ABS. I use the following specification for portfolio shares of fund j in year-quarter r in tranche t of auto ABS deal b issued by i:

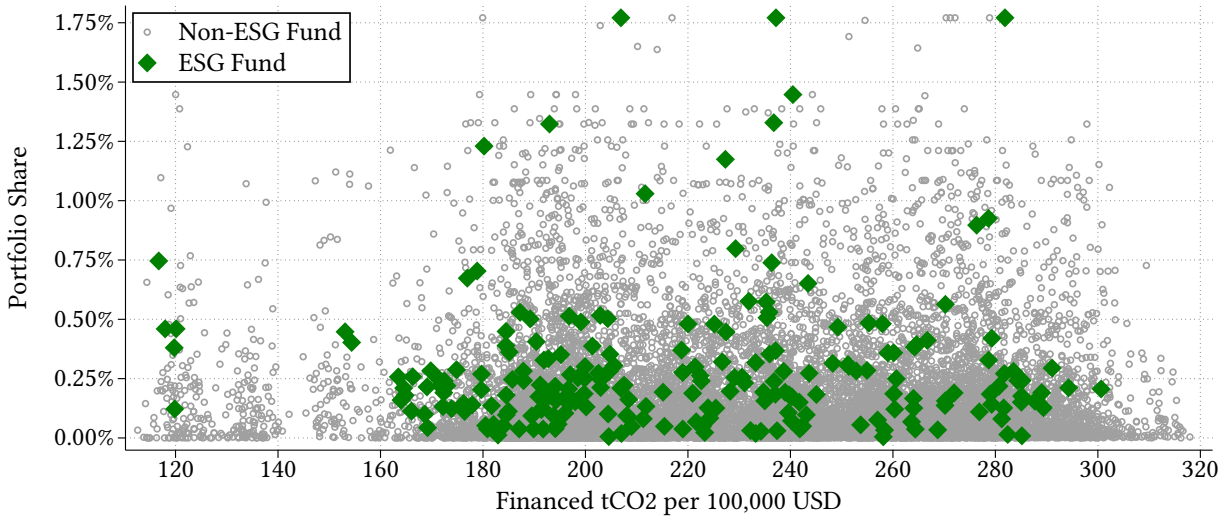
$$\log\left(\text{Portfolio Share}\right)_{jtrb} = \alpha\left(\text{ESG Fund}_j \times \text{Green}_b\right) + \gamma_j + \gamma_b + \gamma_{i\times r} + \mathbf{X}_t'\zeta + \varepsilon_{jtrb} \tag{13}$$

in which Green_b is either a measure of environmental impact such as tCO_2 per vehicle, a measure of energy efficiency such as MPG, or the ESG score of the issuer; γ_j are fund fixed effects; γ_b are auto ABS deal fixed effects; and $\gamma_{i\times r}$ are issuer by reporting year-quarter fixed effects. The coefficient of interest, α, measures the preferences for greenness by ESG funds relative to non-ESG funds. The specifications control for the weighted average life, issuance size, and yield in $\mathbf{X}_t$.

I estimate ESG fund preferences using variation in greenness across multiple auto ABS held by ESG and non-ESG funds during the same period. The specifications include fixed effects for the collateral pool and fund, thus absorbing the characteristics and preferences of each fund and the specific features of each auto ABS. This approach identifies the difference in preference for green assets between ESG and non-ESG funds while controlling for as much unobserved heterogeneity across collateral pools and funds as possible. Additionally, the specifications include issuer by period fixed effects that absorb time-varying issuer characteristics (e.g., issuer health).

Results Figure A2 plots mutual fund portfolio shares in auto ABS against financed CO_2 emissions per $100,000. The graph shows that ESG mutual funds hold positions across the full distribution of CO_2 emissions. This is surprising since common ESG strategies typically involve either outright exclusions of brown assets or best-in-class investments. However, Figure A2 shows that

Figure A2: Portfolio Shares of Mutual Funds in Auto ABS



Notes: This figure shows portfolio shares of ESG and non-ESG mutual funds in auto ABS from 2019-Q3 to 2022-Q2. Portfolio shares are winsorized at 1% and 99%. X-axis is jittered with normally distributed noise for readability.

ESG funds hold similar or higher shares in auto ABS with high-emission intensity.

Table A12 reports estimates of the relationship between greenness and ESG ownership using Eq. (13). The coefficients in Column (1) of Panel A indicate that the greenest 50% of auto ABS receive 20.2% less capital from ESG funds than from non-ESG funds. Columns (2) to (6) present similar estimates using other measures of greenness, all showing positive coefficients of similar magnitude. For example, moving from the 10th to the 90th percentile of average financed CO_2 per vehicle (moving from Honda to Ford) results in a 0.4 standard deviation higher portfolio share for ESG funds than for non-ESG funds.

Panel B of Table A12 repeats the reduced form demand estimation of Panel A but controls for the ESG scores of auto ABS issuers. The estimates in columns (1) and (4) show that both the S&P and Refinitiv ESG scores correlate strongly with the differential demand by ESG funds. Columns (2), (3), (5), and (6) show that controlling for ESG scores shrinks the coefficients on CO_2 emissions toward zero and renders them mostly insignificant.

In summary, ESG funds invest more in auto ABS from issuers with high ESG scores than non-ESG funds. The positive correlation between ESG scores and CO_2 emissions of the collateral means that ESG funds inadvertently invest more in high-emission auto ABS compared with non-ESG funds.[31]

[31] Appendix Table B1 shows the positive correlation of ESG scores and CO_2 emissions in the mutual fund data.

Table A12: Reduced Form Asset Demand System of Mutual Funds for Auto ABS

Panel A: Measures of Environmental Impact of Investment

	(1) Portfolio Share	(2) Portfolio Share	(3) Portfolio Share	(4) Portfolio Share	(5) Portfolio Share	(6) Portfolio Share
ESG Fund=1 × Green (tCO2<p50)=1	-0.226*					
	(0.095)					
ESG Fund=1 × Financed tCO2 per USD		0.154*				
		(0.069)				
ESG Fund=1 × Financed tCO2 per Vehicle			0.144**			
			(0.044)			
ESG Fund=1 × Avg. MPG ×(-1)				0.196***		
				(0.052)		
ESG Fund=1 × Truck Share					0.236*	
					(0.113)	
ESG Fund=1 × Avg. GHG Rating (KBRA)×(-1)						0.202**
						(0.063)
Fund FE	✓	✓	✓	✓	✓	✓
ABS Deal FE	✓	✓	✓	✓	✓	✓
Issuer × Year-Quarter FE	✓	✓	✓	✓	✓	✓
Tranche FE, Tranche controls	✓	✓	✓	✓	✓	✓
Adj. R^2	0.822	0.822	0.822	0.822	0.820	0.819
Observations	11,334	11,334	11,334	11,334	10,919	10,559

Panel B: ESG Scores versus Environmental Impact of Investment

	(1) Portfolio Share	(2) Portfolio Share	(3) Portfolio Share	(4) Portfolio Share	(5) Portfolio Share	(6) Portfolio Share
ESG Fund=1 × Refinitiv ESG Score	0.157**	0.145*	0.115+			
	(0.060)	(0.059)	(0.066)			
ESG Fund=1 × S&P ESG Score				0.112*	0.102+	0.064
				(0.054)	(0.054)	(0.056)
ESG Fund=1 × Financed tCO2 per USD		0.107			0.120	
		(0.084)			(0.088)	
ESG Fund=1 × Financed tCO2 per Vehicle			0.086			0.120+
			(0.066)			(0.066)
Fund FE	✓	✓	✓	✓	✓	✓
ABS Deal FE	✓	✓	✓	✓	✓	✓
Issuer × Year-Quarter FE	✓	✓	✓	✓	✓	✓
Tranche FE, Tranche controls	✓	✓	✓	✓	✓	✓
Adj. R^2	0.821	0.821	0.821	0.821	0.821	0.821
Observations	10,111	10,111	10,111	10,111	10,111	10,111

Notes: This table reports coefficient estimates of Eq. (13). The sample is from 2019-Q3 to 2022-Q2. Coefficients are standardized to unit variances. MPG and GHG Rating are multiplied by (-1) such that higher values are environmentally worse. Standard errors in parentheses are clustered at the fund level. * p<0.10, ** p<0.05, *** p<0.01

B Online Appendix (for online publication only)

(a) Santander SDRIVE 2021-4 Subprime Issue

Pricing $1.8bn Santander Drive Auto Receivables Trust 2021-4

Issuer: Santander Consumer USA
Lead Managers: Citi(str.), JPM, and SIS
DE&I Co-managers: AmeriVet Securities, Great Pacific Securities, Mischler Financial Group

Anticipated Capital Structure:

CL	OFF. AMT	WAL	F/M	L.FNL	BENCH	SPRD	YLD%	CPN	PX
A-1	$222.40	0.13	F1+/P-1	11/15/2022	Intl.	7	0.16802	0.16802	100.00000
A-2	$543.10	0.64	AAA/Aaa	08/15/2024	EDSF	20	0.380	0.37	99.99378
A-3	$292.37	1.43	AAA/Aaa	08/15/2025	EDSF	15	0.517	0.51	99.99081
B	$288.61	2.10	AA/Aaa	06/15/2026	IntS	30	0.887	0.88	99.98887
C	$243.48	2.80	A/Aa2	02/16/2027	IntS	47	1.271	1.26	99.97902
D	$241.38	3.58	BBB/Baa2	10/15/2027	IntS	70	1.685	1.67	99.98569
E	$131.18	3.97	NR/B2	<<NOT OFFERED>>					

(b) CarMax 2019-1 Prime Issue

$1.5bln CarMax (CARMX) 2019-1
JOINT BOOKRUNNERS : Credit Suisse (str), Barclays, Wells Fargo
CO-MANAGERS : MUFG, Scotia, SMBC, TD

CLS	$AMT(MM)	WAL	S&P/FITCH	P.WIN	L.FNL	BNCH	SPRD	YLD%
A1	277.000	0.28	A-1+/F1+	1-7	01/2020	IntL -	1	2.78007
A2A	412.000	1.16	AAA/AAA	7-22	07/2022	EDSF +	31	3.045
A2B	100.000	1.16	AAA/AAA	7-22	07/2022	1mL +	31	
A3	493.900	2.64	AAA/AAA	22-43	03/2024	IntS +	40	3.074
A4	107.910	3.84	AAA/AAA	43-48	08/2024	IntS +	65	3.283
B	42.170	3.98	AA/AA	48-48	11/2024	IntS +	85	3.479
C	39.910	3.98	A/A	48-48	01/2025	IntS +	115	3.779
D	27.110	3.98	BBB/BBB	48-48	08/2025	IntS +	145	4.079

* Exp. Settle: 01/23/19
* First Pay Date: 02/15/19
* Px Speed: 1.30% ABS to 10% Call
* Timing: PRICED
* Format: Public/SEC
* ERISA: Yes
* Min Denoms: $5k by $1k
* B&D: Credit Suisse

Figure B1: Examples of Typical Auto Loan Securitizations

Table B1: Correlation of Greenness Measures in Mutual Fund Holdings

	ESG Score	Env. Score	Financed tCO2/Vehicle	Financed tCO2/USD	Avg. MPG×(-1)	Truck %	GHG Rating×(-1)
ESG Score	1.00						
Environmental Score	0.90	1.00					
Fin. tCO2/Vehicle	0.39	0.14	1.00				
Fin. tCO2/USD	0.31	0.08	0.48	1.00			
Avg. MPG ×(-1)	0.21	-0.01	0.85	0.40	1.00		
Truck %	0.24	0.06	0.83	0.24	0.89	1.00	
GHG Rating ×(-1)	0.13	-0.04	0.78	0.19	0.87	0.90	1.00

Notes: This table reports Spearman rank correlation coefficients across variables in the mutual fund portfolio data. MPG and GHG Rating are multiplied by (-1) such that higher values are environmentally worse.

B.1 Indicator Variable and Matching Estimator

Table B2: Semi-Elasticity with Respect to High-ESG or Low-Emission Indicator

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
			Issuance Spread					
High Refinitiv ESG (score>p50)	-0.108***	-0.082**						
	(0.030)	(0.027)						
High S&P ESG (score>p50)			-0.090+	-0.085+				
			(0.050)	(0.051)				
Low Emissions (USD<p50)					0.047	0.074*		
					(0.032)	(0.028)		
Low Emissions (Vehicle<p50)							0.055*	0.043
							(0.026)	(0.026)
Year-month FE, daily market controls	✓	✓	✓	✓	✓	✓	✓	✓
Prepayment speed FE, tranche controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-ante prepayment controls	✓	✓	✓	✓	✓	✓	✓	✓
Ex-post prepayment controls		✓		✓		✓		✓
Adj. R^2	0.953	0.959	0.950	0.957	0.946	0.953	0.947	0.952
Observations	235	235	235	235	276	276	276	276

Notes: All control variables are in logs. Standard errors in parentheses are clustered at year-month and originator. * p<0.10, ** p<0.05, *** p<0.01

Table B3: Semi-Elasticity Estimates using Propensity Score Matching

	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread
Low Emissions (tCO2<p50)	0.236***		
	(0.0616)		
Top-ESG (Refinitiv Score>p50)		-0.136*	
		(0.0590)	
Top-ESG (S&P Score>p50)			-0.128*
			(0.0563)
Time, Subprime, APS FE	Yes	Yes	Yes
Observations	84	174	198
Treated	50	93	77
Control	34	81	121
# Nearest Neighbors	2	2	2

+ p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Appendix Table B3 shows that one obtains qualitatively similar results to the main results when using a propensity score matching estimator. The "treated" (i.e., either low CO_2 emissions or high ESG score) and "untreated" auto ABS are matched to their $k = 2$nearest neighbors.

B.2 Double-selection Lasso Estimator

Table B4: Estimates using Double-selection Lasso Estimator of Belloni et al. (2014)

	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread	(4) Issuance Spread	(5) Issuance Spread	(6) Issuance Spread
Refinitiv ESG Score	-0.511***	-0.379***	-0.374***			
	(0.0672)	(0.0879)	(0.0817)			
S&P ESG Score				-0.168***	-0.163***	-0.149***
				(0.0356)	(0.0424)	(0.0430)
Financed tCO2 per USD	-0.0729	-0.128	-0.102	-0.208+	-0.119	-0.194*
	(0.113)	(0.115)	(0.0855)	(0.111)	(0.114)	(0.0982)
Time, Subprime, APS, Tranche FE	Yes	Yes	Yes	Yes	Yes	Yes
No. of potential controls	38	290	858	38	290	858
No. of selected controls	11	15	15	11	17	15

All variables in logs. Standard error clustered at year-month. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Appendix Table B4 shows that one obtains qualitatively and quantitatively similar results to the main results when using the double-selection Lasso estimator of Belloni et al. (2014).

The list of potential control variables for the Lasso algorithm is the following: Level of VIX at issuance, standard deviation of the VIX in the 30 days before issuance, inflation expectations (5-Year breakeven inflation rate) at issuance, 6-month and 12-month estimates of the treasury yield curve from Filipović et al. (2025), attachment point, weighted average life of tranche, issuance size of tranche, total issuance size, 30+ days delinquency rate, difference from the assumed prepayment speed, average share of used cars, average interest rate of loans, average warehousing time, 25th percentile of warehousing time, 75th percentile of warehousing time, average credit score of borrowers, 25th percentile of credit score of borrowers, 75th percentile of credit score of borrowers, average loan-to-value ratio at issuance, 25th percentile of loan-to-value ratio at issuance, 75th percentile of loan-to-value ratio at issuance, average % of principal outstanding at the time of securitization, 25th percentile of % of principal outstanding at the time of securitization, 75th percentile of % of principal outstanding at the time of securitization, average remaining term, 25th percentile of remaining term, 75th percentile of remaining term, average original term, 25th percentile of original term, 75th percentile of original term, average vehicle value at origination, 25th percentile of value at origination, 75th percentile of value at origination, captive FE, U.S. issuer FE, as well as interaction terms among these variables. I require the following fixed effects to be present in each (Lasso) regression: assumed absolute prepayment speed, year-month, tranche, and subprime fixed effects.